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Sumo Robot

by

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Mechatronic Project 478

Final Report

Study leader: Mnr W. Swart

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EXECUTIVE SUMMARY

Title of Project
Design and Development of an Autonomous Mini Sumo Robot.
Objectives
Design and construct an autonomous Mini Sumo robot capable of detecting, pushing and removing an opponent from the sumo arena while remaining within the competition boundaries and complying with all competition constraints.
What is current practice and what are its limitations?
Current Mini Sumo robots primarily rely on simple sensor-based control systems for line detection and opponent detection. Their performance is often limited by restricted adaptability to different opponents, inefficient pushing strategies and the strict size, weight and power limitations.
What is new in this project?
This project investigates improved control strategies, sensor integration and mechanical design to enhance the robot's ability to detect opponents, adapt to them, remain within the arena and generate sufficient traction and pushing force during competition.
If the project is successful, how will it make a difference?
The completed robot will demonstrate a balanced integration of sensing, control and mechanical design for competitive Mini Sumo robotics. The resulting platform, control strategies and design methodology can serve as a reference for future robotics competitions. Several subsystems, including the control software and sensor integration techniques, can also be adapted for use in other robotic applications.
What are the risks to the project being a success? Why is it expected to be successful?
Potential risks include unreliable sensor measurements, insufficient wheel traction or pushing force, electronic hazards and software or control issues affecting the robot's behaviour. These risks are mitigated through iterative testing, careful component selection, simulation, prototype evaluation and systematic software development throughout the project to ensure that it is successfull.
What contributions have/will other students make?
The project is conducted independently and does not directly build upon previous university robotics projects. The current work is saturated by hobbyists and competitive teams.
Which aspects of the project will carry on after completion and why?
The robot platform and control software can be further developed to improve autonomous strategies and sensing capabilities. The mechanical performance can also be improved for future robotics competitions or research projects.
What arrangements have been/will be made to expedite continuation?
Comprehensive documentation of the design process, hardware selection, software implementation, testing procedures and experimental results will be discussed to assist future students in extending or improving the project.

Graduate Attribute Summary

GA1. Problem solving

The ability to identify, analyze and solve a complex engineering problem was demonstrated throughout this report, previous reports and oral presentations. Section 2.1 explains why this topic is an open-ended problem and how there is not one correct solution. The problem was analysed and defined throughout Section 1 to Section 3. Section 4 and Section 5 introduce possible approaches and finalizes the solution through both qualitative and quantitative analysis techniques. Furthermore, problem-solving skills were applied throughout testing and tuning as well. For example, the LM2596 converters caused voltage spikes under impact - this problem was solved using linear regulators instead (Chapter 5). During edge-detection, the momentum of the robot carried it over the arena border. This was solved by implementing and fine-tuning a reverse brake (Chapter 7).

Sections	1, 2, 3, 4, 5, 6, 7

GA2. Application of scientific and engineering knowledge

The application of scientific and engineering knowledge can be allocated to past modules. The application of E-design 314 principles can be seen throughout the electronic design (Section 5). Specifically with the integration of microcontrollers, sensors and communication protocols. Electronics 315 was applied in sections regarding circuit safety (diodes, resistors, current, fuses and junction temperature). Computer systems 245 can mainly be seen in Section 6 through the implementation of software, register management and communication protocols. Control systems 354 was used to model the system and apply control theory principles. Machine Design 314 was mainly used for torque and speed determination, the mechanical design of the robot (screw placements, supporting structures and safety factors) and mould design.

Sections	5, 6, Appendix E

GA3. Engineering design

The engineering design process was followed throughout the project, from problem definition to final solution. This included conceptual design, detailed design, prototyping, testing, and iteration to ensure the robot met all functional and technical requirements.

Sections	3, 4, 5, 6, 7, Appendices B, F, G

GA5. Use of engineering tools

Multiple engineering tools were used throughout the design and testing phases (Section 4, Section 5 and Section 7). These tools include Autodesk Fusion (Section 5) for CAD modeling, Autodesk Inventor (Section 5) for FEA analysis and technical drawings, MATLAB (Section 5) for data analysis and simulations, KiCAD and LTSpice (Section 5) for simulations, MATLAB (Section 7) for further data analysis and comparisons of results and ESP-IDF and PlatformIO for software development and debugging (Section 6).

Sections	4, 5, 6, 7, Appendix E

GA6. Professional and technical communication

Professional and technical communication was demonstrated through written reports, oral presentations and documentation of the design process. This includes the clear presentation of technical information, proper referencing of sources, and effective communication of design decisions and results. Additionally, this report adheres to the page limits set per category.

GA8. Individual and collaborative work

While this project mostly involved individual work, collaboration was still neccessary for certain aspects. Such as shared lab resources and space, the process of obtaining components and materials, and seeking guidance from supervisors and peers.

GA9. Independent learning ability

The ability to learn independently was demonstrated through the research and acquisition of knowledge - also known as the literature review (Section 2). Furthermore, new technologies were used to support the project.

Sections	2, 5, 6

Acknowledgements

First of all, I would like to express my gratitude to my supervisor, Mr. Swart, for his guidance and support throughout this project. I would also like to thank my mom for her support and encouragement throughout this project, my dad for letting me 'borrow' his tools and my partner for providing a constant supply of coffee. Lastly, I would like to thank my friends for the company they offered - especially during those late nights in the labs and study centers.

Contents

Plagiarism Declaration **i**

Executive Summary **iii**

Graduate Attribute Summary **iv**

Acknowledgements **vi**

Contents **vii**

List of Figures **xii**

List of Tables **xiv**

List of Symbols **xvi**

1 Introduction **1**
1.1 Background 1
1.2 Objectives 2
1.3 Motivation 2
1.4 Use of Artificial Intelligence 2

2 Literature Review **3**
2.1 An Open-ended Problem 3
2.2 Control Strategies 3
2.2.1 Combat and Adaptive Strategies 3
2.2.2 Search and Attack 3
2.2.3 Boundary Avoidance 4
2.2.4 Fuzzy Logic Control 4
2.2.5 Sensor Arrays in Competitive Robots 4
2.2.6 Latency and Field-of-View Trade-offs 5
2.2.7 Handling Blind Spots 5

2.2.8 Opponent Detection Algorithms . . . 5
2.3 Physical Design . . . 6
2.3.1 Wheel Material . . . 6
2.3.2 Power Supply . . . 6
2.4 Summary . . . 7

3 Specifications and Requirements 8
3.1 Stakeholder Requirements . . . 8
3.2 Functional Requirements . . . 9
3.3 Technical Performance Measures . . . 10
3.4 Conclusion . . . 10

4 Concept Design 12
4.1 Strategy Approach . . . 12
4.2 Movement . . . 13
4.2.1 Concept One: Four-Wheel Drive . . . 13
4.2.2 Concept Two: Two-Wheel Drive . . . 13
4.2.3 Concept Evaluation . . . 14
4.3 Opponent detection . . . 14
4.3.1 Concept One: IR Sensors . . . 15
4.3.2 Concept Two: ToF Laser Sensors . . . 15
4.3.3 Concept Three: Ultrasonic Sensors . . . 15
4.3.4 Concept Evaluation . . . 15
4.4 Engage opponent . . . 16
4.4.1 Concept One: Flat Wedge . . . 16
4.4.2 Concept Two: Curved Wedge . . . 16
4.4.3 Concept Three: Forked Wedge . . . 17
4.4.4 Concept Evaluation . . . 18
4.5 Control and measure motion . . . 18
4.5.1 Concept One: Light-interrupter Wheel Encoders . . . 18
4.5.2 Concept Two: Magnetic (Hall-effect) Wheel Encoders . . . 18
4.5.3 Concept Three: IR-sensor Wheel Encoders . . . 19
4.5.4 Concept Four: IMU . . . 19
4.5.5 Concept Evaluation . . . 19

5 Design Refinement 20
5.1 Electronic Components . . . 20
5.1.1 Motor Selection . . . 20
5.1.2 Motor Driver Selection . . . 21
5.1.3 Microcontroller Selection . . . 21
5.1.4 Edge Detection Sensors . . . 22

5.1.5 Power Supply . . . 22
5.1.6 Voltage Regulator and Circuit Protection . . . 23
5.1.7 Switch and Push-button . . . 25
5.1.8 I2C Communication . . . 25
5.2 Mechanical Components . . . 26
5.2.1 Sensor Placement . . . 27
5.2.2 Wedge . . . 28
5.2.3 Side Frame . . . 29
5.2.4 Chassis . . . 29
5.2.5 Wheels . . . 30
5.2.6 Electronic mounts . . . 31

6 Software Design and implementation 33
6.1 Software Overview and Requirements . . . 33
6.2 Task Structure and Scheduling . . . 33
6.3 System Model . . . 34
6.4 Feedback and Control System . . . 35
6.5 Behaviour State Machine . . . 35
6.5.1 Start-up Sequence . . . 36
6.5.1.1 Initialization and Calibration . . . 36
6.5.1.2 Match Start and Countdown . . . 36
6.5.2 Dodge Algorithm . . . 37
6.5.3 Search Algorithm . . . 37
6.5.3.1 Target Detection . . . 37
6.5.3.2 Alignment . . . 37
6.5.4 Attack Algorithm . . . 38
6.5.4.1 Approach . . . 38
6.5.4.2 Ram and Impact Detection . . . 39
6.5.4.3 Push and Woodpecker . . . 39
6.5.5 Back-off Algorithm . . . 40
6.5.5.1 Edge Detection . . . 40
6.5.5.2 Edge Recovery Manoeuvre . . . 40
6.5.6 Telemetry . . . 40

7 Testing and Refinement 41
7.1 PID Controller Application . . . 41
7.1.1 Experimental Setup and Procedure . . . 41
7.1.2 Experimental Results . . . 42
7.2 Pivot and Turning Performance . . . 42
7.2.1 Experimental Setup and Procedure . . . 43
7.2.2 Experimental Results . . . 43

7.3 Maximum Robot Speed . . . 44
7.3.1 Experimental Setup and Procedure . . . 44
7.3.2 Experimental Results . . . 44
7.4 Dohyo Edge Detection Response . . . 45
7.4.1 Experimental Setup and Procedure . . . 45
7.4.2 Experimental Results . . . 45
7.5 Opponent Detection . . . 46
7.5.1 Experimental Setup and Procedure . . . 46
7.5.2 Experimental Results . . . 46
7.6 Pushing Force . . . 47
7.6.1 Experimental Setup and Procedure . . . 47
7.6.2 Experimental Results . . . 47
7.7 Additional Tests . . . 48
7.8 Autonomous Matches . . . 49

8 Conclusions and Future Work 50

A Final Built Sumo Robot 51
A.1 Competition Restraints: Size and Weight . . . 51

B Evaluation Criterion 52
B.1 Concept Design Evaluation . . . 52
B.2 Refined Design Evaluation . . . 54

C Testing 55
C.1 VL53L0x Testing . . . 55
C.2 QRE1113 Sensor Testing . . . 56
C.3 Friction Tests . . . 57
C.4 Turning Tests . . . 57
C.5 Heading Tests . . . 58

D Techno-Economic Analysis 60
D.1 Project Schedule . . . 60
D.2 Budget . . . 61
D.3 Technical Impact . . . 62
D.4 Return on Investment . . . 63
D.5 Potential for Commercialisation . . . 63
D.6 Responsible Use of Resources and End-of-Life Strategy . . . 63

E Calculations and Simulations 64

F Software Flow Diagrams **67**
F.1 Task Structure Overview 67
F.2 State Machine Overview 68
F.3 Initialization flow 69

G Electrical Schematic **70**

H Safety Report and Risk Assessment **71**

List of References **76**

List of Figures

1.1.1 Standard mini sumo competition arena (Parallax Inc., 2005). 1

2.2.1 Neuro-Fuzzy Controller Block Diagram (Kamal *et al.*, 2018). 4

4.2.1 Four-wheel drive concept. 13
4.2.2 Two-wheel differential drive concept. 14
4.4.1 Conceptual views of the flat-wedge design. 16
4.4.2 Conceptual views of the curved-wedge design. 17
4.4.3 Conceptual views of the forked-wedge design. 17

5.1.1 Electronic Components Overview. 20
5.1.2 Power Supply Schematic 24
5.1.3 Main Power Switch and Start Button 25
5.1.4 I2C Communication Setup 25
5.1.5 VL53L0x XShut Pin Configuration 26
5.2.1 Final CAD assembly of the robot. 27
5.2.2 Sensor reading test 27
5.2.3 Sensor placement simulation 28
5.2.4 FEA Analysis of the wedge design. 29
5.2.5 Final side panel design. 29
5.2.6 Final chassis design. 30
5.2.7 Wheel hub and mould designs. 31
5.2.8 Electronic mounts for the components within the frame. 31
5.2.9 Full electronic layout plan. 32

6.3.1 Robot kinematics (California Institute of Technology, 2018). 34
6.4.1 PID Controller Step Response Simulation 35
6.5.1 Overview of the behaviour state machine. 36
6.5.2 Fuzzy logic for alignment 38

7.1.1 Heading without PID Controller 42
7.1.2 PID Controller Results 42

7.2.1 Response to a 90° step at 170 DC 43
7.2.2 Response to a 90° step at 250 DC 43
7.4.1 Edge detection response with (a) the motors stopped and (b) a momentary brake. 45

8.0.1 The Final Built Sumo Robot ('Jeremy') 50

A.1.1 Final built sumo robot weight. 51
A.1.2 Final built sumo robot size. 51
A.1.3 Final built sumo robot height. 51

C.1.1 VL53L0x Preliminary Testing Setup 55
C.4.1 Turning Test Setup 57
C.4.2 Turning Test Telemetry Results 58
C.5.1 Heading Test Telemetry Results 59

D.1.1 Original Planned Project Schedule. 60
D.1.2 Revised Project Schedule. 60

E.0.1 Motor-related Calculations. 64
E.0.2 Battery-related Calculations. 65
E.0.3 Heatsink Calculations. 65
E.0.4 Friction Test Results for Wheel Materials. 66

F.1.1 Task Structure Overview 67
F.2.1 State Machine Overview 68
F.3.1 Initialization Flow Diagram 69

G.0.1 Electrical Schematic 70

List of Tables

2.4.1 Summary of the literature review 7

3.1.1 Stakeholder requirements 8
3.2.1 Functional requirements for the autonomous sumorobot 9
3.3.1 Technical performance measures for the autonomous sumorobot . . . 10

4.0.1 Morphological chart of the design concepts 12
4.2.1 Drive-train concept evaluation based on FRs 14
4.3.1 Opponent detection concept evaluation based on FRs 15
4.4.1 Wedge concept evaluation based on FRs 18
4.5.1 Motion measurement concept evaluation based on FRs 19

5.1.1 Qualitative motor concept evaluation 21
5.1.2 Comparison of microcontrollers 22
5.1.3 Comparison of battery technologies 23

6.2.1 FreeRTOS task priorities and core assignments 33

7.2.1 Pivot turn step response results 44
7.3.1 Maximum forward speed results at full duty (255) 44
7.4.1 Edge detection response 46
7.5.1 Opponent detection reliability at different search spin speeds 47
7.6.1 Pushing test results at full duty (255) 47

B.1.1 Evaluation criteria and their justification 52
B.2.1 Comparison of opponent detection technologies 54
B.2.2 Qualitative comparison of PLA, PETG and TPU filaments 54
B.2.3 Comparison of wheel materials 54

C.1.1 VL53L0X measured distance (cm) versus actual distance and incidence angle 56
C.2.1 QRE1113 edge detection under varying light conditions 56
C.3.1 Incline angle at which each wheel material slipped 57

D.2.1 Theoretical cost of engineering and workshop time 61
D.2.2 Actual project costs. 62

List of Symbols

Constants

g = $9.81\,\mathrm{m/s^2}$

Variables

x	Position along the x-axis	[m]
y	Position along the y-axis	[m]
$\dot{x}$	Velocity in the x-direction	[m/s]
$\dot{y}$	Velocity in the y-direction	[m/s]
θ	Robot orientation (heading)	[rad]
$\dot{\theta}$	Rate of change of orientation	[rad/s]
θ_{FOV}	Sensor field of view	[°]
R_a	Armature resistance	[Ω]
$\omega_{\max}$	Maximum search spin rate	[°/s]
T_j	Junction temperature	[°C]
T_a	Ambient temperature	[°C]
$R_{\theta JA}$	Junction-to-ambient thermal resistance	[°C/W]
$R_{\theta JC}$	Junction-to-case thermal resistance	[°C/W]
$R_{\theta CS}$	Case-to-sink thermal resistance	[°C/W]
r	Wheel radius	[m]
L	Distance between the two wheels	[m]
u_l	Left wheel velocity	[rad/s]
u_r	Right wheel velocity	[rad/s]
u	Action vector / controller output	[−]
$u(t)$	PID controller output	[−]
$e(t)$	Error signal	[−]
t	Time	[s]
τ	Integration variable (time)	[s]
K_p	Proportional gain	[−]

K_i	Integral gain	[−]
K_d	Derivative gain	[−]
T	Motor torque	[N·m]
V	Applied motor voltage	[V]
K_t	Motor torque constant	[N·m/A]
K_e	Back-emf constant	[V·s/rad]
ω	Angular velocity	[rad/s]
t_{meas}	Sensor measurement time	[s]
n	Number of consecutive readings to confirm a target	[−]
P_{diss}	Power dissipated by the component	[W]
f	Forward speed command (out of 255)	[−]
s	Closeness of the target (0 to 1)	[−]
μ_{fL}	Front-left sensor closeness value	[−]
μ_{fR}	Front-right sensor closeness value	[−]

Subscripts

a	Ambient / armature
d	Derivative
e	Back-emf
i	Integral
j	Junction
l	Left
r	Right
p	Proportional
t	Torque
fL	Front-left
fR	Front-right
max	Maximum
meas	Measurement
diss	Dissipated

Chapter 1

Introduction

1.1 Background

The design of a Mini Sumo-robot is mainly limited by the competition guidelines and rules. This includes the physical dimensions of the robot, the characteristics and layout of the competition arena and the operational requirements and limits that must be satisfied during a match. The Mini sumobot class is restricted to a maximum footprint of 100 mm x 100 mm with no height limit (RoboGames, 2026). The weight is limited to 500 g as well and the robot must be fully autonomous - with no remote control or human intervention allowed during a match.
The Robogames mini sumo competition arena, also known as the dohyo, is a circular platform with a diameter of 77 cm and a height of 2.5 cm. The surface is made of a smooth, black melamine coated wood and the boundary is marked by a white border with a width of 2.5 cm as seen in Figure 1.1.1. Inside this arena, two centerlines (also called shikiri-lines) are drawn to indicate the starting positions of the robots.

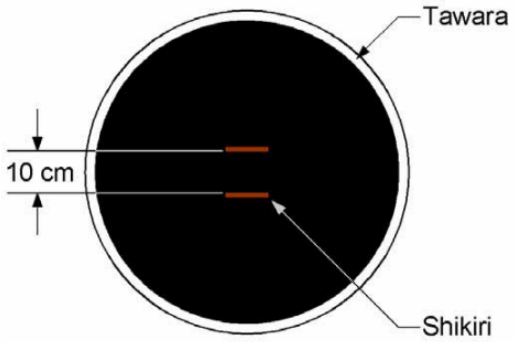


Figure 1.1.1: Standard mini sumo competition arena (Parallax Inc., 2005).

The main objective of these robots is to push the opponent out of the dohyo while avoiding being pushed out themselves. A 'Yuko' is scored when a robot is pushed out of the dohyo and the first robot to score two 'Yuko' wins the match. It should be noted that the use of any jammers or devices that could cause harm to the opponent is strictly prohibited.

1.2 Objectives

The objective of this project is to design and develop a compact autonomous Mini Sumo-robot that complies with the Robogames Mini Sumo competition regulations. The robot must be capable of detecting opponent robots while remaining within the arena boundaries. A suitable control strategy will be developed to enable the robot to autonomously locate, engage and push its opponent out of the arena while avoiding the boundary line. Finally, the completed system will be validated through experimental testing and participation in the Mini Sumo knock-out competition to demonstrate its functionality and overall performance.

1.3 Motivation

Autonomous robotics combines theoretical knowledge, design and testing in a loop of trial, error and improvement. The project integrates mechanical design, the design of electronic systems and the implementation of control and strategy algorithms. This project makes use of knowledge gained in the Computer systems 245, Electronics 315, Control systems 354, E-design 314 and Machine Design 314 modules. By integrating these areas, the project provides an opportunity to demonstrate the practical application of engineering principles in the design and development of an autonomous robotic system.

1.4 Use of Artificial Intelligence

The use of artificial intelligence in this report was strictly limited to an overall spell and grammar check. All non-referenced content, including diagrams and figures, were created by the author without AI assistance.

Chapter 2

Literature Review

2.1 An Open-ended Problem

Autonomous sumo robots must detect, engage and push an opponent without any human intervention once a match has started (RoboGames, 2026). Although the Mini Sumo rules limit the size, mass and autonomy of the robot, they do not provide specific guidance on the design. Furthermore, the opponent's design and strategy are unknown beforehand. The project is therefore open-ended as there is no single correct solution and the design must be justified through testing and iteration. It is also a complex engineering problem. The 500 g mass and 100 mm $\times$ 100 mm footprint limits force trade-offs between conflicting requirements such as pushing force, endurance and agility. The project combines mechanical, electronic and software design to reach a solution.

2.2 Control Strategies

2.2.1 Combat and Adaptive Strategies

Competitive sumo robots generally combine a set of tactics: a start-up manoeuvre, a search pattern, a committed attack and a boundary-avoidance reflex (Vera Vera *et al.*, 2024; RoboGames, 2026). The mechanical design also influences the strategy, including a low centre of mass and a plough or wedge at the front that helps the robot get under the opponent and resist being pushed (Zaręba *et al.*, 2022).

2.2.2 Search and Attack

When no opponent is detected, the robot must search the dohyo. Common approaches are to rotate on the spot while scanning with the front sensors or to drive in a curved path around the ring. Once the opponent is detected, the robot commits to an attack. If the opponent leaves the sensing range, the robot returns to its search behaviour (Vera Vera *et al.*, 2024).

2.2.3 Boundary Avoidance

When the white border is detected, the robot reverses or turns away from the edge before resuming the search or attack. This is usually given the highest priority since leaving the ring forfeits the match (RoboGames, 2026).

2.2.4 Fuzzy Logic Control

Fuzzy logic control is a method that uses some degrees of freedom to determine the output rather than a binary true/false approach. It is particularly useful for systems with uncertainty or imprecision such as those involving complex environmental interactions (Pantanetti *et al.*, 2025). Fuzzy logic has therefore been used to track the opponent smoothly instead of through binary decisions (Pantanetti *et al.*, 2025). One approach uses a fuzzy inference system for detecting and tracking the opponent in the ring with a neural network used to extract rules and tune the fuzzy parameters (Erdem, 2011). Artificial intelligence has also been identified as a route to improving detection, manoeuvres and decision making, but it requires more computing resources than a simple state machine (Vera Vera *et al.*, 2024).

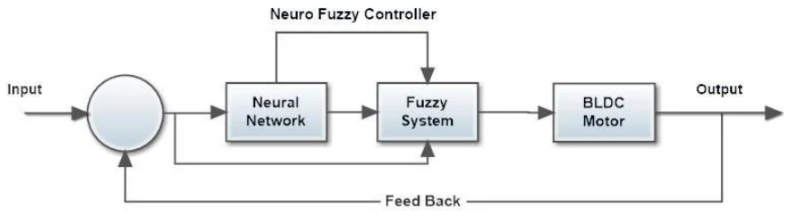


Figure 2.2.1: Neuro-Fuzzy Controller Block Diagram (Kamal *et al.*, 2018).

2.2.5 Sensor Arrays in Competitive Robots

Competitive sumo robots rarely rely on a single sensor. Instead, they use arrays of IR or time-of-flight sensors that cover different directions. The densest coverage is towards the front where the opponent is most likely going to be. For example, one open-source mini sumo design uses fourteen VL53L0X time-of-flight sensors of which eight give coverage around the whole robot. Six give finer resolution in the forward direction (OliviliK, n.d.). Analogue IR designs follow the same principle. The "Number Two" mini sumo robot uses ten modulated IR emitters and eight detectors to locate an opponent in any direction, with most of the IR output concentrated at the front (Robot Room, n.d.).

2.2.6 Latency and Field-of-View Trade-offs

The choice between sensor types is a trade-off between response time and information content. IR detectors respond within microseconds but only indicate that something is present. Time-of-flight sensors provide a distance measurement but need tens of milliseconds per reading (Table B.2.1).
Field of view creates a second trade-off. A narrow field of view allows the direction of the opponent to be estimated accurately, but a larger number of sensors is needed to avoid gaps between their cones. A wide field of view needs fewer sensors but gives a coarse bearing so the robot has to turn further to align itself with the opponent. For IR emitters and detectors, the output must also be tuned. If it is too weak, a dark opponent is missed at long range and if it is too strong, the ring or spectators may be detected (Robot Room, n.d.).

2.2.7 Handling Blind Spots

Gaps between the sensor cones and sensors that are shadowed by the robot's own wedge or body create blind spots. Top designs reduce these by combining a wide array of sensors with a denser forward array (OliviliK, n.d.) and by physically shielding the detectors so that each one only responds to its own direction. In the "Number Two" robot, black foam between the detectors prevented side and rear reflections from triggering other detectors (Robot Room, n.d.). When the opponent is not detected at all, the robot's search routine (Section 2.2.1) compensates by rotating until the opponent enters a sensor's field of view. This is fast but struggles with ambiguous readings which has motivated the use of fuzzy logic.

2.2.8 Opponent Detection Algorithms

The simplest detection algorithm is a threshold on each sensor with the robot turning towards whichever sensor is triggered. Recently, an embedded fuzzy logic sumo robot using analogue IR distance sensors and an Arduino Mega was proposed to deal with the ambiguity of sensor readings that rule-based if-else logic handles poorly (Mohd Shukri *et al.*, 2025). These approaches remain rare compared with the threshold-based logic used by most competitive robots, since they add computation and tuning effort for a limited gain in a task that is mainly dominated by reaction speed. Camera-based systems are less common in mini sumo due to their higher computational requirements and slower response times compared with IR or time-of-flight sensors.

2.3 Physical Design

2.3.1 Wheel Material

Polyurethane (PU) is commonly used for wheels because of its high abrasion and tear resistance. Commercially available polyurethane elastomers range from roughly 25–90 Shore A with densities of 1.24–1.27 g/cm^3 (Zenith Rubber Europe, 2021). Higher Shore hardness reduces deformation while increasing resistance to wear. This makes polyurethane suitable for applications requiring durable and high-load wheels.
Silicone rubber is a softer elastomer that is also available for various Shore A hardness values. Silicone products are commonly available from 25–80 Shore A (Revchem Composites, 2022). Its lower hardness allows for greater conformity to the contact surface. This increases the contact area but also increase rolling deformation. While silicone rubber breaks down less easily than polyurethane, it is much more expensive. Additionally, it does not stick to surfaces during the moulding process and does not require a release agent.
Styrene-Ethylene-Butylene-Styrene (SEBS) is a thermoplastic elastomer commonly used for wheels because it combines rubber-like flexibility with good abrasion and tear resistance. SEBS is available in a wide range of hardnesses, from approximately 10–80 Shore A, with densities ranging from roughly 0.87–0.93 g/cm^3 for unfilled grades (Mexpolimeros, 2025).

2.3.2 Power Supply

The battery selection of an autonomous sumo robot affects its mass and operating time. Common battery technologies include lithium polymer (LiPo), lithium-ion (Li-ion) and alkaline batteries.
LiPo batteries have a nominal cell voltage of 3.7 V and can provide high discharge currents due to their low internal resistance. Their specific energy is typically approximately 130–200 Wh/kg, while discharge ratings can exceed 20C for high-performance cells (Battery University, 2025). LiPo batteries are therefore commonly used in applications requiring a high power to battery mass ratio.
Li-ion batteries have a nominal cell voltage of roughly 3.6–3.7 V and a specific energy of approximately 150–250 Wh/kg (Battery University, 2025). They generally provide higher energy density than LiPo cells but standard Li-ion cells have lower maximum discharge currents.
Alkaline batteries have a nominal voltage of 1.5 V per cell and a specific energy of approximately 80–110 Wh/kg (Battery University, 2025). Their internal resistance increases as the battery discharges, limiting their ability to provide sustained high currents. They can therefore be used for low-power electronics but are less suitable

for applications requiring high instantaneous power.

2.4 Summary

Table 2.4.1 summarises the prior solutions reviewed in this chapter and their limitations.

Table 2.4.1: Summary of the literature review

Topic	**Existing solutions**	**Limitations**
Combat strategy	Start-up manoeuvre, search, committed attack and boundary-avoidance reflex (Vera Vera *et al.*, 2024).	Fixed state machines are predictable and can be exploited.
Fuzzy and adaptive control	Fuzzy inference for smooth opponent tracking (Pantanetti *et al.*, 2025); neuro-fuzzy tuning (Erdem, 2011; Kamal *et al.*, 2018); embedded fuzzy sumo robot (Mohd Shukri *et al.*, 2025).	Extra computation and tuning effort; AI needs more resources than a small microcontroller offers (Vera Vera *et al.*, 2024).
Opponent sensing	Arrays of IR or ToF sensors, densest at the front (OliviliK, n.d.; Robot Room, n.d.).	IR is fast but gives no distance; ToF gives distance but needs tens of ms per reading; blind spots between sensor cones.
Wedge and mechanical design	Low centre of mass and a front wedge or plough to get under the opponent (Zaręba *et al.*, 2022).	Wedge shape affects both engagement and where sensors can be mounted.
Wheel materials	Polyurethane, silicone rubber and SEBS (Zenith Rubber Europe, 2021; Revchem Composites, 2022; Mexpolimeros, 2025).	Datasheet properties do not give the real friction on the dohyo.
Power supply	LiPo, Li-ion and alkaline cells (Battery University, 2025).	Alkaline and standard Li-ion cells cannot supply high motor currents.

Chapter 3

Specifications and Requirements

3.1 Stakeholder Requirements

The stakeholder requirements for the autonomous sumorobot are listed in Table 3.1.1. The requirements are ranked in order of importance, with the most important requirement being ranked 'i'. The primary stakeholders for this project include that staff and students who are directly involved in the manufacturing and testing as well as the final competition of the robot. The requirements are therefore based on the needs of these stakeholders alongside the competition rules and regulations.

Table 3.1.1: Stakeholder requirements

SR ID	Description	Rank
SR1	The robot must comply with the Mini-Sumo competition rules.[1]	i
SR2	The robot must operate autonomously during the match.	ii
SR3	The robot must remain within the dohyo boundary.	iii
SR4	The robot must have enough traction and pushing force to compete against an opponent.	iv
SR5	The robot must withstand impacts or collisions and pushing forces without mechanical or electrical failure.	v
SR6	The robot must operate reliably throughout testing and competition without unexpected resets or system-related failures.	vi
SR7	The robot must complete a full match using its contained power supply.	vii

Continued on next page

[1]This includes the applicable size, mass, starting delay, autonomous operation and competition requirements.

Table 3.1.1 continued from previous page

SR ID	Description	Rank
SR8	The robot must be manufacturable using the available facilities, components and project budget.	viii

3.2 Functional Requirements

The functional requirements are derived from the stakeholder requirements and are listed in Table 3.2.1 below.

Table 3.2.1: Functional requirements for the autonomous sumorobot

FR ID	Description	Derived from
FR_01	The robot shall detect the opposing robot.	SR2
FR_02	The robot shall detect the edge of the dohyo.	SR3
FR_03	The robot shall move within the dohyo.	SR2
FR_04	The robot shall apply sufficient force to push the opposing robot out of the dohyo.	SR4
FR_05	The robot shall respond to the opposing robot in real time.	SR2, SR6
FR_06	The robot shall operate autonomously without human intervention after the start command.	SR1, SR2, SR3
FR_07	The robot shall adhere to the size and weight constraints of the competition.	SR1
FR_08	The robot shall be self-contained and shall not rely on external power, communication or physical support during a match.	SR1, SR7
FR_09	The robot shall begin operation only after the required competition start delay.	SR1
FR_10	The robot shall not damage the dohyo, the opposing robot or itself during normal operation.	SR1, SR5
FR_11	The robot shall be manufactured from materials and designed in such a way that it can withstand the forces and impacts experienced during a match.	SR5

3.3 Technical Performance Measures

The technical performance measures (TPMs) are derived from the functional requirements. These performance measures will be used to evaluate and test the system.

Table 3.3.1: Technical performance measures for the autonomous sumorobot

TPM ID	**Related FR**	**Description**	**Target**	**Range**	**Unit**
TPM_01	FR_09	Start delay after power-on[1].	5	5–10	s
TPM_02	FR_01, FR_05	Opponent detection time[6].	–	0–100	ms
TPM_03	FR_07	Robot footprint[1].	100×100	$\leq 100 \times 100$	mm
TPM_04	FR_07	Robot mass[1].	500	≤ 500	g
TPM_05	FR_01, FR_05	Detection angle of sensor [10].	30	20–40	°
TPM_06	FR_02, FR_05	Dohyo edge detection time[9].	20	0–30	ms
TPM_07	FR_03	Maximum forward speed[7].	0.30	0.10–0.77	m/s
TPM_08	FR_04	Available wheel torque[2].	1.5	≤ 1.5	kgf·cm
TPM_09	FR_01	Opponent detection range[3].	770	0–770	mm
TPM_10	FR_04	Continuous pushing duration[5].	5	2–10	s
TPM_11	FR_08	Continuous operating time[4].	9	>3	min
TPM_12	FR_03	Steady-state heading error[8].	≤ 5	≤ 10	°
TPM_13	FR_03	Overshoot[8].	≤ 2	≤ 10	%
TPM_14	FR_03	Settling time[8].	≤ 300	≤ 500	ms
TPM_15	FR_03	Rise time[8].	≤ 200	≤ 300	ms
TPM_16	FR_05	Control loop latency[8].	≤ 10	≤ 20	ms

3.4 Conclusion

Most of the measurable functional requirements have been addressed through the specified technical performance measures and goals. However, some regarding

the autonomous nature and safe design of the robot are not linked to specific measurable values as they were considered more qualitative in nature. They were still included as they were used to guide certain design decisions in Chapter 4 and Chapter 5.

[1] According to the Robogames Mini Sumo competition rules.
[2] Wheel torque is specified per drive motor at the gearbox output shaft (see Appendix E).
[3] The sensor must be able to detect the opponent within the specified arena (770 mm).
[4] Matches last for 3 minutes maximum according to the Robogames Mini Sumo competition rules. Ideally, the robot should be able to run for 3 consecutive matches without recharging or replacing the power source.
[5] Continuous pushing duration is based on the average time it takes to push an opponent out of the dohyo or to overcome wheel slip. The motor should not stall within this time.
[6] Estimate sensor processing time based on typical operation ratings for various sensors.
[7] Reaching this speed would allow the robot to cross roughly half of the dohyo in a second.
[8] These values are based on the expected performance of the robot during a match. Faster response is prioritized over perfect accuracy.
[9] At maximum speed of roughly 30 cm/s the robot will move 6 mm before detecting or reacting to the edge (20 ms x 30 cm/s = 6 mm).
[10] Ideal detection angle based on Section 2.2.6 and existing sensor specifications (see Table B.2.1).

Chapter 4

Concept Design

Based on previous approaches, discussed in Section 2, an autonomous sumorobot can essentially be broken down into four sub-systems: the drive-train, the opponent sensing system, the engagement mechanism and the control system. All other components of the robot, such as the chassis, power supply and other sensors will depend on the design of these four sub-systems. The morphological chart in Table 4.0.1 summarizes the different design options considered for each function in this section.

Table 4.0.1: Morphological chart of the design concepts

		Options			
Functions	**Drive-train**	Four-wheel	Two-wheel	–	–
	Opponent sensing	IR sensors	ToF laser sensors	Ultrasonic sensors	–
	Engage opponent	Flat wedge	Curved wedge	Forked wedge	–
	Motion control	Light-interrupter encoders	Hall-effect wheel encoders	IR-sensor wheel encoders	IMU

4.1 Strategy Approach

In many competitive sumo robots the strategy is selected before the match using a DIP switch or similar manual input. This allows the team to adapt to a known opponent, but the strategy remains fixed once the match has started. This approach heavily depends on the operator's prediction of how the opponent will behave.

In this project, the aim is to automate the behaviour of the robot completely. The robot will instead infer the opponent's behaviour from its sensor readings during the match. This keeps the robot fully autonomous and allows it to cope with opponents whose designs and tactics are unknown beforehand. This goal is discussed here as a means to highlight why certain design decisions were made throughout this project.

4.2 Movement

4.2.1 Concept One: Four-Wheel Drive

The four-wheel drive concept uses four independently driven wheels with each wheel controlled by its own motor. Driving all four wheels increases traction, stability and pushing force while distributing the robot's weight more evenly across four contact points. The additional motors increase mass, power consumption, wiring complexity and the space required within the 100 mm $\times$ 100 mm size limit. The four-wheel layout also experiences greater turning resistance due to wheel skidding.

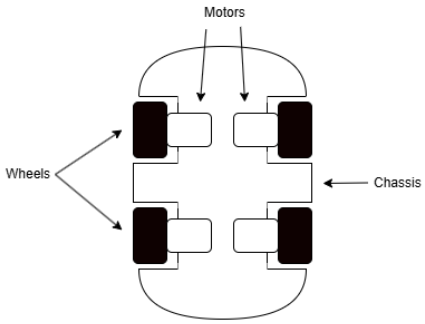


Figure 4.2.1: Four-wheel drive concept.

4.2.2 Concept Two: Two-Wheel Drive

The two-wheel setup is commonly used to ease control of movement and turning.Differential drive is known for its zero turning radius (Sisko, 2017), which allows the robot to turn in place and quickly change direction. While this design won't reach the same level of traction and pushing force as a four-wheel drive, it is simpler to implement and is commonly used when the goal is to out-maneuver the opponent rather than out-torque them.

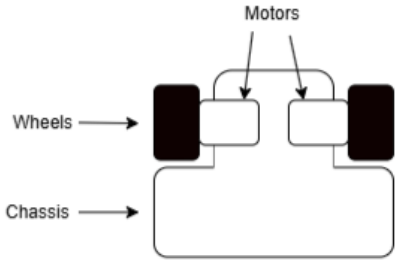


Figure 4.2.2: Two-wheel differential drive concept.

4.2.3 Concept Evaluation

The drive-train concepts were evaluated using criteria based on the functional requirements of the sumobot. The decision matrix shown in Table 4.2.1 was used to compare these two concepts and further explanation for these criteria can be found in Table B.1.1. The weights total up to 1 and the values are assigned on a scale of 0 to 2, where 0 is the worst and 2 is the best.

Table 4.2.1: Drive-train concept evaluation based on FRs

Criterion	**Linked FR**	**Weight**	**4WD**	**2WD**
Manoeuvrability	FR_03	0.35	1	2
Traction	FR_04	0.35	2	1
Complexity	FR_07	0.1	1	2
Integration	FR_07	0.2	1	2
Weighted score		**1**	**1.35**	**1.65**

The two-wheel drive concept was selected with a score of 1.65 compared to the four-wheel drive's score of 1.35.

4.3 Opponent detection

The robot must detect the opponent (FR_01) before it can respond (FR_05). The target has an unknown shape and material, must be seen up to 770 mm (TPM_09) and must be found within 100 ms (TPM_02). Three sensor types were considered: IR, ToF laser and ultrasonic sensors.

4.3.1 Concept One: IR Sensors

IR sensors emit infrared light and measure how much is reflected back. Obstacle modules have a range of 2–80 cm and a 35° beam. They are simple and fast, but they cannot measure distance reliably and are affected by ambient light and by the colour of the target (Sisko, 2017).

4.3.2 Concept Two: ToF Laser Sensors

ToF sensors time the return of a light pulse. Most commonly used modules measure 30 mm to 2 m, run directly from 3.3 V and are less sensitive to ambient light and target angle than IR (STMicroelectronics, 2024). Their 20–33 ms measurement time is within TPM_02.

4.3.3 Concept Three: Ultrasonic Sensors

Common ultrasonic modules like the HC-SR04 times the echo of a 40 kHz pulse with a range of 2–400 cm and a 15–20° beam (ELECFreaks, n.d.; ArcBotics, n.d.). A full-range echo takes about 23 ms. However, it struggles with angled and soft targets as these will redirect the pulse away from the receiver.

4.3.4 Concept Evaluation

Alongside the quantitative chart, a qualitative assessment was also conducted using locally available sensor modules for each category. This assessment can be seen in Table B.2.1. The quantitative evaluation is summarised in Table 4.3.1.

Table 4.3.1: Opponent detection concept evaluation based on FRs

Criterion	**Linked FR**	**Weight**	**IR**	**ToF**	**Ultrasonic**
Range and coverage	FR_01	0.25	1	2	2
Response time	FR_01, FR_05	0.20	2	1	1
Robustness	FR_01	0.25	0	2	1
Integration	FR_07	0.15	2	1	1
Distance output	FR_01	0.15	0	2	2
Weighted score		**1**	**0.95**	**1.65**	**1.40**

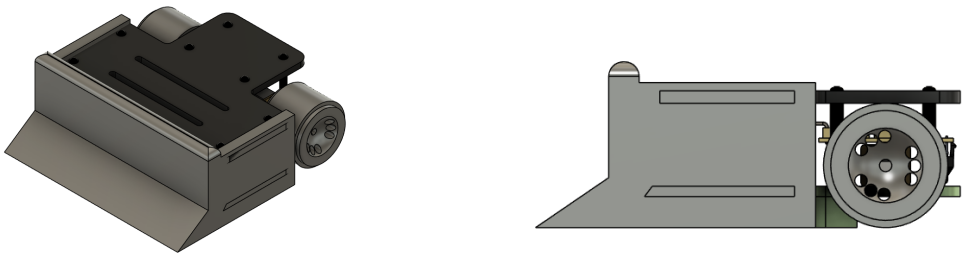
ToF laser sensors were selected with a weighted score of 1.65.

4.4 Engage opponent

Section 2 discusses why mainly wedges are considered as the method of engaging with the opponent. It allows the robot to break the traction of the opponent and redirect its motion.

4.4.1 Concept One: Flat Wedge

The first wedge concept considered was a flat wedge. A shallow, flat wedge design can slide under the front edge of the opponent and lift it, which reduces the load on the opponent's wheels and cuts its traction. Its main weakness is that the straight face gives little sideways control, so a robot with a taller or stronger wedge can instead push the flat wedge up and over and a lifted opponent can still be pushed off at an angle.

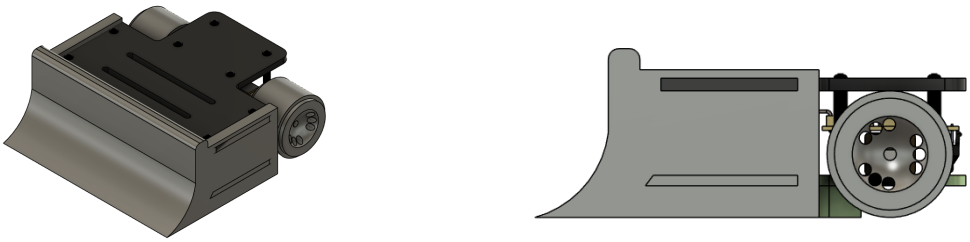


(a) Isometric view (b) Side view

Figure 4.4.1: Conceptual views of the flat-wedge design.

4.4.2 Concept Two: Curved Wedge

The curved wedge design is similar to the flat wedge, but with a curved front edge translating into a straight flat surface. This design is intended to help the robot redirect the opponent's wedge instead of lifting it. The curve turns the contact force upward and sideways, so the opponent is deflected away from the robot rather than lifted. It works best against opponents with their own wedge, whose motion it can redirect and it is less sensitive to the exact height of the contact. Its main weakness is that the curved front surface lifts less than a flat wedge.



(a) Isometric view (b) Side view

Figure 4.4.2: Conceptual views of the curved-wedge design.

4.4.3 Concept Three: Forked Wedge

The two prongs straddle the opponent's wedge and lift it from both sides which should give a more stable lifting action while leaving a flat surface for sensors between the prongs. It works best against opponents with a single central wedge or a narrow front. Its main weakness is narrow contact since each prong carries the load on its own and doesn't allow as much contact surface with the opponent.



(a) Isometric view (b) Front top-down view

Figure 4.4.3: Conceptual views of the forked-wedge design.

4.4.4 Concept Evaluation

Table 4.4.1: Wedge concept evaluation based on FRs

Criterion	Linked FR	Weight	Flat	Curved	Forked
Force redirection	FR_04	0.45	1	2	1
Engagement	FR_04	0.2	1	1	2
Sensor integration	FR_01	0.1	2	1	2
Internal space	FR_07	0.15	2	1	1
Manufacturability	SR8	0.1	2	1	0
Weighted score		**1**	**1.35**	**1.45**	**1.2**

The curved wedge was selected with a score of 1.45 compared to 1.35 for the flat wedge and 1.2 for the forked wedge.

4.5 Control and measure motion

To keep the robot moving in a straight line and turn by a controlled amount (FR_03), the control system needs feedback.

4.5.1 Concept One: Light-interrupter Wheel Encoders

A light-interrupter encoder uses a slotted disc that passes between an LED and a photodetector, producing a pulse each time a slot passes. It gives accurate speed readings with digital outputs, but it requires a disc mounted on an extended shaft or a wheel large enough to house the sensor.

4.5.2 Concept Two: Magnetic (Hall-effect) Wheel Encoders

A Hall-effect encoder reads a small magnetic disc on the motor shaft and is commonly available for micro gear motors. It produces the same type of digital pulses as the optical encoder, but it is enclosed and is not affected by dust or ambient light. The disc needs an extended rear shaft.

4.5.3 Concept Three: IR-sensor Wheel Encoders

For most motor options found during research, pre-mounted encoders were not available. Using an IR reflective sensor to detect markings on the wheel was also considered. This allows more flexibility in the mounting, but it needs a larger wheel radius to house the sensor which increases speed and decreases torque. Its output is also affected by ambient light.

4.5.4 Concept Four: IMU

The IMU combines a three-axis gyroscope and accelerometer on a small board. Its gyroscope measures the rotation rate of the robot directly, so the heading can be found by integrating it and is not affected by wheel slip.

4.5.5 Concept Evaluation

Table 4.5.1: Motion measurement concept evaluation based on FRs

Criterion	**Linked FR**	**Weight**	**Optical**	**Hall**	**IR**	**IMU**
Heading measurement	FR_03	0.30	1	1	1	2
Wheel speed measurement	FR_03	0.20	2	2	1	0
Integration	FR_07	0.20	0	1	0	2
Robustness	FR_03, FR_11	0.15	0	1	0	1
Interface (pins)	–	0.15	1	1	0	2
Weighted score		**1**	**0.85**	**1.20**	**0.50**	**1.45**

The IMU was selected with a score of 1.45, ahead of the Hall-effect encoder (1.20). Encoders only give heading indirectly, through the difference between the wheel speeds, and this is corrupted by wheel slip whenever the robot is pushing. The IMU measures the heading directly, so the controller can correct any speed mismatch between the two motors by adjusting their relative speeds.

Chapter 5

Design Refinement

While Section 4 discussed the four main sub-systems of the sumo robot, this chapter focuses on refining the entire system design. Various solutions were considered for some of these refinements and a qualitative evaluation was performed to select the most suitable options.

5.1 Electronic Components

The electronic section discusses the operating requirements and selection of the electronic components for the sumo robot. Figure 5.1.1 provides an overview of the electronic sub-components and which section discusses which component.

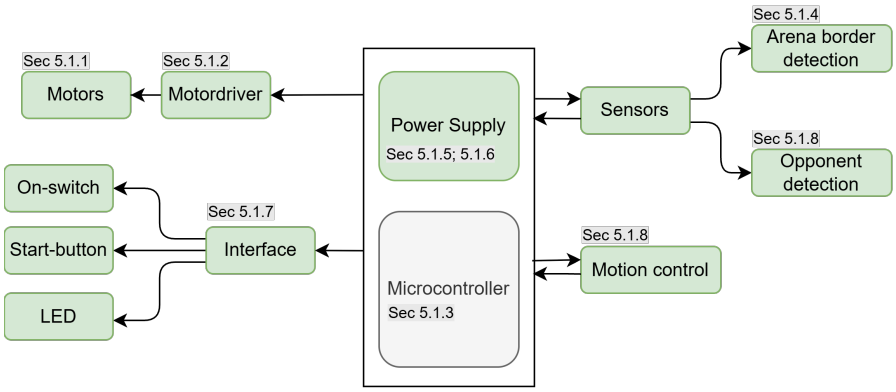


Figure 5.1.1: Electronic Components Overview.

5.1.1 Motor Selection

The motors had to fit within the 10×10 cm footprint alongside the wheels and battery and, more importantly, be locally available. Most small high-torque motors are made for American or European sumo competitions. Base on locally available options, brushed and brushless DC motors were compared as seen in

Table 5.1.1. Brushless motors are commutated electronically which gives them higher efficiency, power density and lifespan (Scholten, 2025). However, they need a three-phase driver and commutation in software. Brushed motors are commutated mechanically and can be driven directly by an H-bridge with a PWM signal.

Table 5.1.1: Qualitative motor concept evaluation

Criterion	**Brushed DC**	**Brushless DC**
Speed and torque	Moderate, lower torque	Higher power to weight density
Local availability	High	Low
Driver complexity	Low	High
Size and mass	Small and light	Moderate paired with larger driver

A brushed 12 V, 300 rpm N20 metal gear motor was selected. Its rated speed of 280 rpm (Robo Factory, n.d.) gives a top speed of 0.46 m/s with a 30 mm wheel - theoretically meeting TPM_07 (See Appendix E, Figure E.0.1 . The trade-off is a rated torque of only 1 kgf·cm for both motors. However, the stall torque of 4 kgf·cm per motor (Robo Factory, n.d.) leaves some headroom.

5.1.2 Motor Driver Selection

A TB67H420FTG Dual motor driver carrier was selected to control the motors. This motor driver is capable of driving two DC motors with a supply voltage of 10V to 47V and is rated to supply 1.7 A continuously per motor (Micro Robotics, 2026*a*). This makes it suitable for the 12 V N20 motors which are rated to draw 1.2 A at stall - allowing for a safety factor of 1.4 (see Figure E.0.2). The motordriver uses MOSFETS whereas older motordrivers utilize Bipolar Junction Transistors (BJTs). MOSFETS are more efficient than BJTs, which means that they can handle higher currents and dissipate less heat. Additionally, the carrier was chosen as it has built-in protection against under-voltage, over-current and over-temperature conditions (Micro Robotics, 2026*a*).

5.1.3 Microcontroller Selection

The microcontroller is the central processing unit of the autonomous sumorobot. Given its dependence on all other peripherals, the microcontroller could only be selected towards the end of the design process. During the selection process, there were several limiting factors. The microcontroller had to fit within the

10cm × 10cm footprint of the robot while providing sufficient processing capability and enough GPIO pins with the required functions to interface with all of the selected peripherals. The following microcontrollers were considered:

Table 5.1.2: Comparison of microcontrollers

Specification	**ESP32 DevKit**	**Arduino Nano**	**STM32F411 Blackpill**
Processor	240 MHz	16 MHz	100 MHz
Flash	4 MB	32 KB	512 KB
GPIO	25	22	30+
Board Size	51 × 28 mm	45 × 18 mm	52 × 25 mm

The ESP32 DevKit V1 was selected as the most suitable option for the robot. Its relatively small development-board footprint, large number of configurable GPIO pins and support for communication protocols such as I^2C and SPI allow it to interface with all of the required peripherals. The ESP32 provides sufficient processing and peripheral capabilities for the autonomous sumorobot with a dual-core 32-bit processor operating at up to 240 MHz and 4 MB of flash memory (Random Nerd Tutorials, n.d.). Figure G.0.1 shows the pins needed for each peripheral which aids in the planning of the pin configuration for the microcontroller.

5.1.4 Edge Detection Sensors

The main goal of the edge detection sensors are to detect the switch-over from black to white - making IR sensors the most suitable solution. Coupled with the insight that ESPs generally have noisy ADCs, a QRE1113 IR sensor with a digital output was chosen. These sensors operate at a voltage of 3.3 V and are rated to operate at 3 mm from the surface (Micro Robotics, 2026*b*).

5.1.5 Power Supply

Three commonly used options were taken into account when selecting the power supply for the robot (see Section 2.3.2):

Table 5.1.3: Comparison of battery technologies

Parameter	LiPo	Li-ion	Alkaline
Nominal cell voltage	3.7 V	3.6–3.7 V	1.5 V
Specific energy	130–200 Wh/kg	150–250 Wh/kg	80–110 Wh/kg
Discharge capability	High	Moderate–High	Low
Internal resistance	Low	Low–Moderate	High
Rechargeable	Yes	Yes	No

Lithium polymer (LiPo) batteries were chosen due to their high energy density to mass ration. Alongside all of the upsides for using LiPo batteries, it should be noted that they require careful handling and charging to avoid safety hazards (See Appendix H). LiPo batteries can be dangerous if punctured, overcharged or exposed to high temperatures (ChinaHobbyLine, 2024). With all this in mind, a 3S 11.1V LiPo battery with a capacity of 750mAh and a discharge rate of 45C was selected as the power supply for the system. A datasheet could not be found for this specific component - instead, the specifications provided by the supplier were used to determine the suitability of the battery for the system.
Given the 750mAh and 45C rating of the battery, the maximum current that can be drawn from the battery is 33.75A. Using this and using conservative estimates for all present electrical components, the total current draw of the system was calculated to be 4.4A (see Figure E.0.2). This offers a safety margin 8 which is sufficient to ensure that the battery can provide enough current to all components without being overdrawn. Using the same maximum estimate current draw of the system, the run time is 11 minutes with this battery configuration - meeting TPM_11.

5.1.6 Voltage Regulator and Circuit Protection

The system was originally designed with a LM2596 DC-DC buck converter to step down the voltage and regulate the 5 V and 3.3 V lines. However, after running safety tests, it was found that the LM2596 converters were prone to voltage fluctuations and spikes, especially when the board was subjected to physical impacts. This would be a problem given the expected impacts the sumo robot would experience during a match. These converters were ultimately replaced with linear regulators. A L78C5 and LD1117 were used to regulate the 5 V and 3.3 V lines respectively. The circuits were built using an 1N4007 rectifier diode to protect against reverse voltage as well as 100 nF and 10 µF capacitors to reduce noise. A power-indicating LED was connected to the 5 V line to indicate when the board

was powered. Figure 5.1.2 shows the final power supply schematic. A preliminary simulation was conducted in LTspice to verify the configuration. At normal operating voltage, the motordriver would be supplied with 11.49 V after the voltage drop over the diodes and stable 5 V and 3.3 V rail would follow after their respective regulators (see Appendix E, Figure E.0.3).

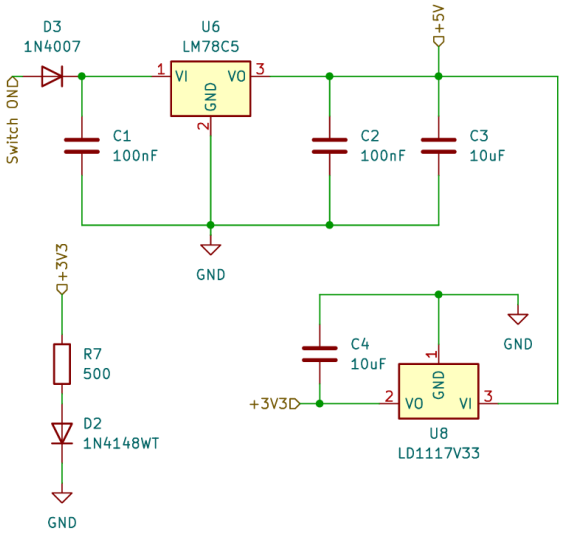


Figure 5.1.2: Power Supply Schematic

Given that this is not a pre-built power supply module, further calculations were conducted to determine the need for a heatsink (See Figure E.0.3). The full equation used to calculate the junction temperature is:

$$T_j = T_a + (R_{\theta JA} + R_{\theta JC} + R_{\theta CS}) \cdot P_{diss} \tag{5.1.1}$$

Where T_j is the junction temperature, T_a is the ambient temperature, $R_{\theta JA}$ is the junction-to-ambient thermal resistance, $R_{\theta JC}$ is the junction-to-case thermal resistance, $R_{\theta CS}$ is the case-to-sink thermal resistance, and P_{diss} is the power dissipated by the component. The junction temperature at estimated operating conditions when the battery is fully charged (12.6 V) was calculated to be 110°C. This allows for a safety margin of only 0.7. Therefore, including a heatsink was the safer option. Alongside the regulated voltage lines, a 300 mA fuse will be installed in series with the power supply to the microcontroller to protect all important components from overcurrent situations. Ideally, the microcontroller should not pull more than 150 mA while its GPIO pins are rated for 8 mA (Espressif Systems, 2026).

5.1.7 Switch and Push-button

The LiPo battery needs to be connected to the system using an XT30 connector. A main power switch is implemented to safely allow for the disconnection and reconnection of the battery. A second push-button is implemented to allow the system to meet TPM_01. The start button is connected to the microcontroller using active-high configuration. A 10 kΩ pull-down resistor ensures that a maximum current of 0.33 mA will flow through the pull-down resistor when the button is pressed, which is well within safe limits for the microcontroller's GPIO pins.

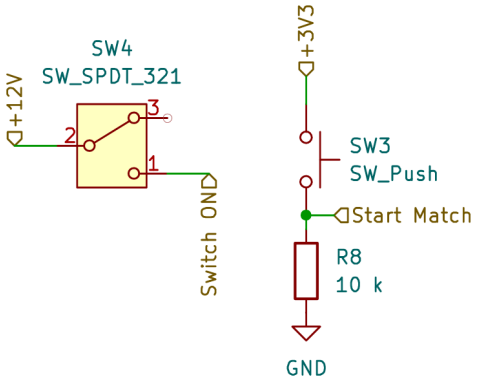


Figure 5.1.3: Main Power Switch and Start Button

5.1.8 I2C Communication

Section 4 discussed the implementation of ToF sensors for opponent detection. The exact sensor chosen was the VL53L0x Time-of-Flight distance sensor which uses I2C communication protocols. For motion control, the LSM6DS3 IMU was chosen as it uses newer technology, has less noise than older models and is configured at exactly 1g. Since both sensors use I2C communication, they share the same bus. The I2C bus allows multiple devices to share the same data and clock lines, reducing the number of GPIO pins required for communication. External pull-up resistors are used on the SDA and SCL lines to ensure faster communication.

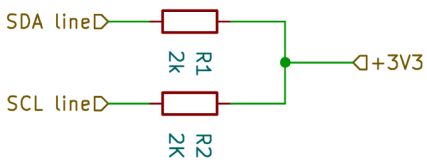


Figure 5.1.4: I2C Communication Setup

For the VL53L0x sensors specifically, the XShut pins are assigned to a GPIO pin on the microcontroller. These sensors operate at a default address of 0x29. When using multiple sensors on the same I2C bus, the XShut pins can be used to assign unique addresses to each sensor.

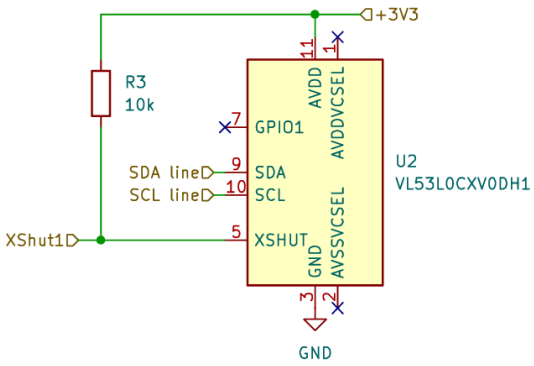


Figure 5.1.5: VL53L0x XShut Pin Configuration

The LSM6DS3 allows for both I2C and SPI communication. To configure it for I2C, the CS pin must be pulled high while the SDO/SA0 pin is connected to ground. Additionally, the readings of the LSM6DS3 IMU can conjest the I2C lines - causing system freezes or missed readings. To avoid this, the INT1 pin of the IMU is connected to an interrupt-capable GPIO pin on the microcontroller, allowing the system to time readings and log faults.

5.2 Mechanical Components

Figure 5.2.1 shows the final CAD assembly of the robot. Given the weight constraints (FR_07), it was chosen to manufacture most of the mechanical components with PETG. 3D printing in general was the prefered method of manufacturing to allow for safer prototyping - mitigating manufacturing risks. Other materials such as PLA and TPU were considered, but PETG was chosen for its balance of strength, ease of printing and thermal resistance (see Table B.2.2).

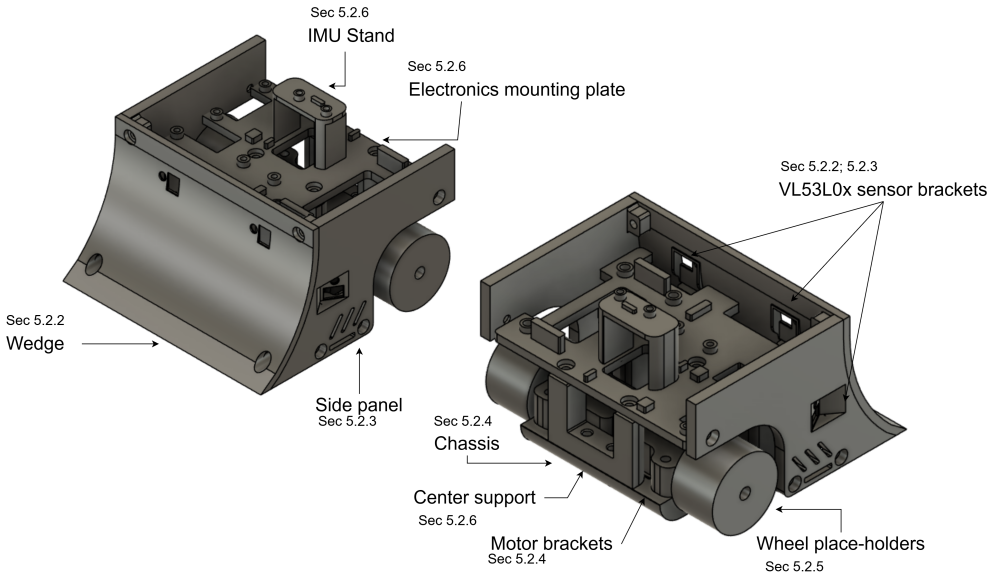


Figure 5.2.1: Final CAD assembly of the robot.

5.2.1 Sensor Placement

A preliminary test was performed regarding the sensor placement and what their optimal placement or orientation might be. The goal was to detect as much of the arena while using as few sensors as possible. Figure 5.2.2 shows the results of the sensor readings and what the actual detection range of the VL53L0X sensors being used is. See C.1 for the test setup and procedure.

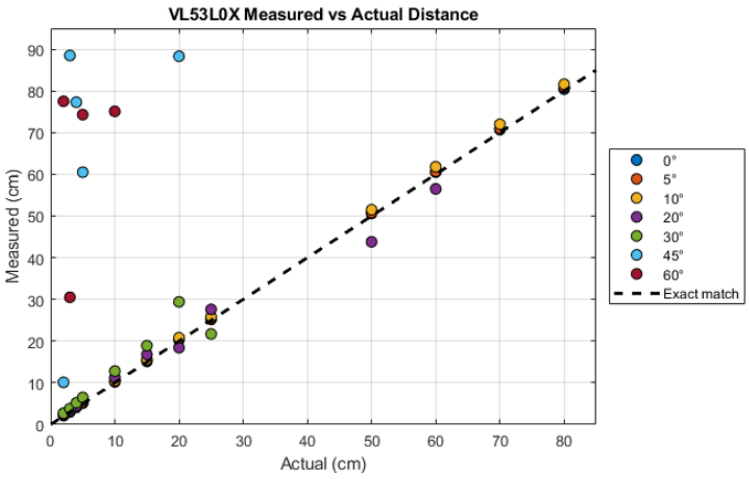


Figure 5.2.2: Sensor reading test

The sensors stopped detecting the test object at an angle of roughly 20–30°, meeting TPM_05. After that, the sensors read the distance of a wall in the background - grouping the readings around 80 cm. This angle was used to plot the different sensor placements in MATLAB - simulating what placement would be the most effective.

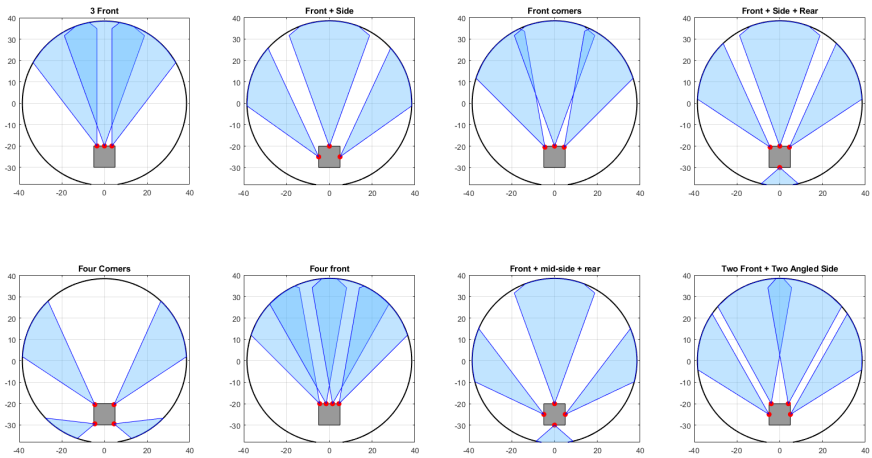
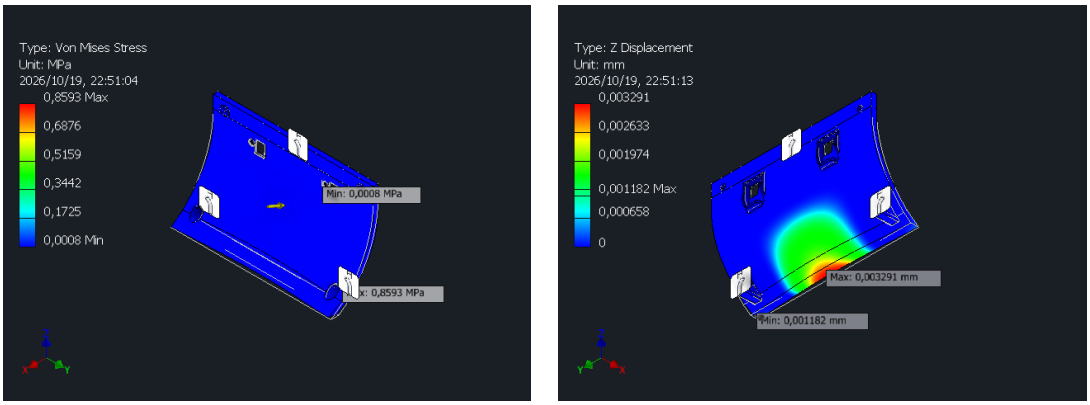


Figure 5.2.3: Sensor placement simulation

Two front facing sensors paired with two angled side-facing sensors was selected as the optimal sensor-array configuration for the robot.

5.2.2 Wedge

While a curved wedge was chosen during the concept development stage, it still needed to be refined to interface with all other components such as the chassis and sensor mounts. A key consideration throughout the entire mechanical design process was the screw placement. All screw placements were angled to align with the expected direction of the forces, minimizing the risk of shearing - especially since the chosen material (PETG) has limited shear strength.
A FEA (Finite Element Analysis) was performed to evaluate the structural integrity of the wedge under anticipated loads - especially given the uncertainty of whether PETG would be able to withstand the forces given the thickness chosen for this part. A load force of 15N was applied - 3 times the estimated impact load of 4.5N (See Figure E.0.1). A much higher load was chosen as the opponent's design and strategy remains a variable. The analysis showed that the wedge could withstand the force and deflection only occured towards the lower part - avoiding electrical components that were placed higher.

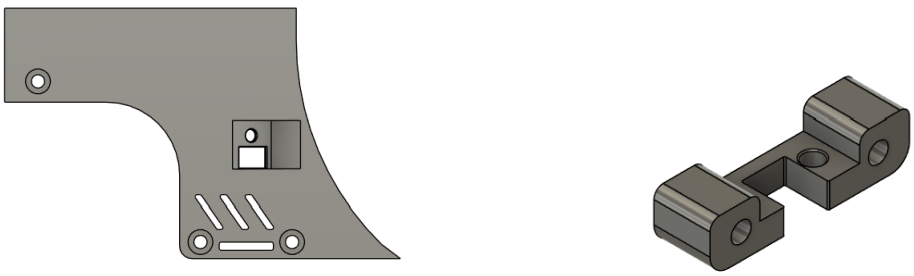


(a) FEA Von Mises stress distribution. (b) FEA displacement.

Figure 5.2.4: FEA Analysis of the wedge design.

5.2.3 Side Frame

The sides are mainly there to provide a mounting surface for the sensors and to protect the electronics from collisions. The sides were designed to mount one VL53L0X sensor alongside a small bracket that mounts the IR edge detection sensor. The diagonal and horizontal cuts near the lower side allow for more natural light to reach the IR sensor.



(a) Front view of the side. (b) IR sensor mount.

Figure 5.2.5: Final side panel design.

5.2.4 Chassis

The chassis consists of mounting space for the motors, a bracket to protect the battery and a slot to place extra weights within. This slot sits directly under the

drive line - allowing the robot to have a lower center of gravity and more stability (as discussed in Section 2). Accompanying the chassis is a lid design to enclose the weights as well as motor-brackets for the N20 motors.



(a) Final chassis. (b) N20 Motor bracket.

Figure 5.2.6: Final chassis design.

5.2.5 Wheels

Three commonly used materials for bot-specific wheels were considered and discussed in Section 2.3.1. While Table B.2.3 was used as a qualitative analysis of these materials and used to guide the initial decision, the true friction factor can only be determined through practical testing.
The friction test was conducted once arena testing could be started. Two Shore-ratings of polyurethane were tested (30 and 40 A). Additionally, store-bought wheels (SEBS) were included in the test for comparison. The test setup and calculations can be found in Figure E.0.4. Shore 30 A polyurethane (Vytaflex) was selected as it has the highest static friction coefficient when tested on the arena. However, this material fuses easily with PETG. While it is ideal for the polyurethane to fuse to the wheel hub, a release agent paired with a draft angle of 2° was used to ensure that the polyurethane would not fuse to the mould cavity.



(a) Wheel mould section view.

(b) Wheel hub top view

Figure 5.2.7: Wheel hub and mould designs.

Initially, there were some concerns about keeping the hub in place or preventing it from floating. The density of PETG is 1.25 g/cm^3 while the density of Vytaflex is 1.15 g/cm^3. Since the density of PETG is higher than that of Vytaflex, the hub would sink into the mold cavity and floating should theoretically not be an issue. The wheel hub was designed in such a way that the polyurethane would slot into the six outer gaps - making it more difficult to release or shear when the robot is in motion. Furthermore, the center slot was designed with two extra gaps around the arched side of the D-slot to allow the N20 motor shaft to be inserted into the hub more easily.

5.2.6 Electronic mounts

Apart from the Vl53L0x and QRE1113 sensors, electronic mounts were designed to keep the components organized and secure within the frame. Circular extrusions facing upwards were added to isolate screws or metal fasteners from wires. Rectangular extrusions were added for stability and cutouts were included to route wires from the chassis or side-frames.

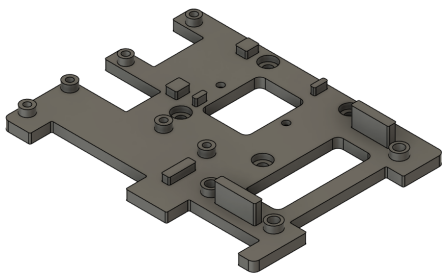


Figure 5.2.8: Electronic mounts for the components within the frame.

The placement of the electronics specifically focused on keeping access points as close as possible as well as ensuring that the I2C lines were kept short to avoid signal degradation. Additionally, a separate mount was designed for the IMU. While the geometry may seem odd, it was designed in such a way that the IMU would be placed directly above the center of the robot's drive-line. The full layout of these mounts can be seen in Figure 5.2.9.

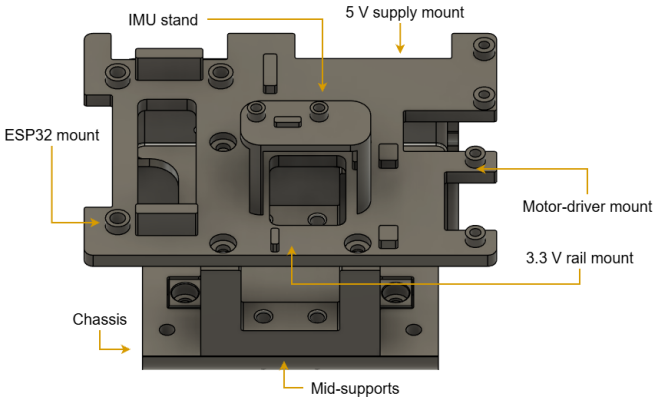


Figure 5.2.9: Full electronic layout plan.

Chapter 6

Software Design and implementation

6.1 Software Overview and Requirements

Given the autonomous nature of the sumo robot, the software had to account for several requirements. The robot must start safely, find the opponent, push it out and never leave the ring itself. To achieve this, the ESP32 microcontroller was programmed using the ESP-IDF framework with FreeRTOS.

6.2 Task Structure and Scheduling

An overview of the task structure is shown in Figure F.1.1. The tasks were split in such a way to avoid stalling, freezing or delaying certain processes. Edge detection was given the highest priority since leaving the arena immediately forfeits the match. To clarify a misleading convention regarding FreeRTOS - a higher number refers to a higher priority. Furthermore, each core has its own scheduling process.

Table 6.2.1: FreeRTOS task priorities and core assignments

Task	**Priority**	**Core**
Edge sensor task	7	0
Sensor task	5	0
Status LED task	5	0
Control task	6	1
Idle task[1]	-	0

[1]The 'Idle task' refers to the initialization sequence before the start-button press.

6.3 System Model

Differential-drive robots can be summarized by a state description consisting of both orientation and position (California Institute of Technology, 2018). This state can be modeled within the x-y plane as shown in Figure 6.3.1.

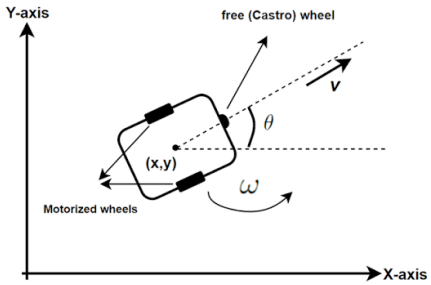


Figure 6.3.1: Robot kinematics (California Institute of Technology, 2018).

Assume the action vector of the robot is represented by $u = [u_r, u_l]$, where u_l and u_r represent the left and right wheel velocities respectively. When both wheels rotate at the same angular velocity ($u_l = u_r > 0$), the differential-drive robot moves in a straight line along its current heading. The resulting linear velocity is directly proportional to the wheel radius, r. When the wheels rotate at equal speeds in opposite directions ($u_l = -u_r \neq 0$), the robot undergoes rotational motion about the midpoint of the axis of the wheels, producing no translational movement. From there on, the following equations can be used to determine the robot's position and orientation over time:

$$\dot{x} = \frac{r}{2}(u_r + u_l)cos(\theta) \tag{6.3.1}$$

$$\dot{y} = \frac{r}{2}(u_r + u_l)sin(\theta) \tag{6.3.2}$$

$$\dot{\theta} = \frac{r}{L}(u_r - u_l) \tag{6.3.3}$$

Where (x, y) is the position of the robot in the x-y plane, θ is the orientation of the robot and L is the distance between the two wheels (Ras, 2025). Furthermore, the individual motors can be modeled as:

$$T = \frac{K_t}{R_a} V - \frac{K_t K_e}{R_a} \omega \tag{6.3.4}$$

Where T is the motor torque, V is the applied voltage, Kt is the motor torque constant, Ke is the back-emf constant, Ra is the armature resistance and ω is the angular velocity (University of Michigan, n.d.).

6.4 Feedback and Control System

A closed-loop control system was implemented in certain states to ensure accurate motion and positioning of the sumo robot. The implemented equation can be displayed as follows:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt} \tag{6.4.1}$$

Where $u(t)$ is the control output, $e(t)$ is the error signal, K_p is the proportional gain, K_i is the integral gain and K_d is the derivative gain. An initial simulation was conducted using the system model discussed in Section 6.3 and the values of K_p, K_i and K_d were adjusted to achieve the desired performance. Using a step response of 90 degrees, a value of $K_p = 16$ and $K_d = 0.6$ were found to provide a good balance between responsiveness and stability.

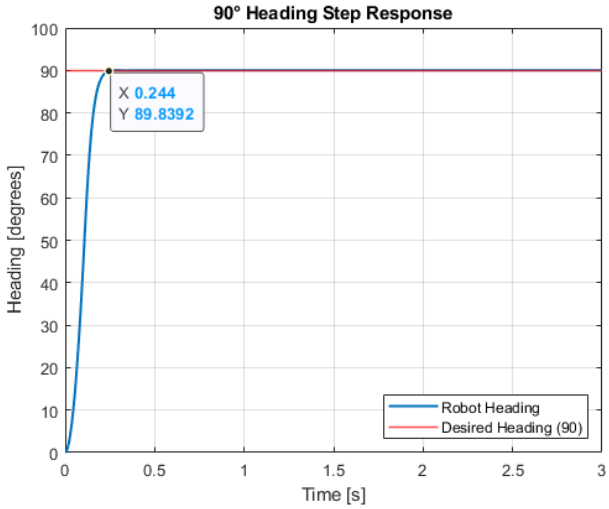


Figure 6.4.1: PID Controller Step Response Simulation

As seen in Figure 6.4.1, the chosen values reach the desired output after 240 ms. This value falls within the acceptable range for TPM_15. However, since the motor parameters were estimated, the response of the robot can only truly be verified during system testing.

6.5 Behaviour State Machine

A finite state machine (FSM) was used to clearly separate states and easily verify transitions between these states. This FSM has a total of fourteen states which

can be classified as the setup, opening, search, attack and safety states. Figure F.2.1 displays the detailed overview of the behaviour state machine while Figure 6.5.1 provides a simplified version for reference throughout this chapter.

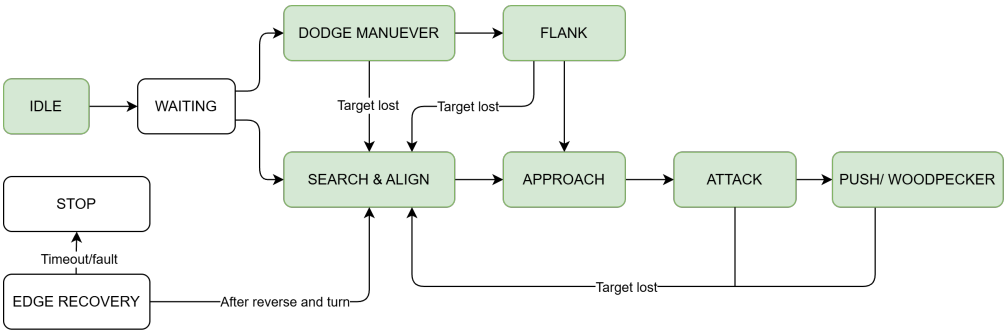


Figure 6.5.1: Overview of the behaviour state machine.

6.5.1 Start-up Sequence

6.5.1.1 Initialization and Calibration

The initialization sequence starts off with the PWM and GPIO pins for motor control. The PWM pins are configured at 20 kHz to keep them above the audible range and set to 8-bit resolution. After the motor control pins are configured, the ToF VL53L0x sensors power up at the same I2C address. By holding their respective XSHUT pins low and releasing them one at a time, each sensor can be assigned a unique address. The IMU is then initialized. On initialization, INT1 is configured for the data-ready interrupt and the current readings of the accelerometer and gyroscope are taken as the 'new zero' reference. It should be noted that the robot can still operate when either the IMU or VL53L0x sensors fail. The on-board LED is used to report these failures beforehand, but a timed run can still continue in the case of needing to test certain peripherals seperately.

6.5.1.2 Match Start and Countdown

The 'start' button is configured as an active-high push button (See Section 5.1.7). Once pressed, a 5-second countdown begins. The Gyro-bias calibration is performed during this countdown while the sumobot is still stationary. After the countdown, the robot resets its heading and PID controllers. If a front sensor already detects the opponent, the robot transitions to the 'STARTUP_WAIT' state. Otherwise, it transitions to the 'SEARCHING' state (See Figure 6.5.1 and Figure F.2.1).

6.5.2 Dodge Algorithm

The dodge algorithm was implemented since most opponents immediately charge at the start of the match. To counter this, the algorithm measures the speed of an approaching opponent within the first second. The speed is calculated based on the change in distance readings from the front VL53L0x sensors over time. If an approach of 500 mm/s is detected, it is treated as a charging opponent. To counter this, the following steps are followed: a turn of 48 to the right with PID control followed by a straight drive for 400 ms, then a left arc (forward 100, turn 65) back towards the opponent's side and finally ALIGNING when the target is reacquired. If no opponent is detected or if the approach speed is below the threshold, the robot continues with the normal search and attack routines (see Figure 6.5.1).

6.5.3 Search Algorithm

6.5.3.1 Target Detection

Given the sensor array discussed in Section 5.2.1, the target detection relies on the readings from the front and side ToF sensors to detect the opponent and determine its relative position. The detection range was originally designed for 770 mm (TPM_09). However, since readings beyond the arena border kept triggering false charges during testing, this was reduced to 300 mm. When no target is visible, the robot is in the SEARCHING state and spins on the spot in a counter-clockwise direction (Section 7.5.1 for the spin duty-cycle). When the opponent is detected within range, the robot considers it a valid target and proceeds with the alignment. The thresholds are summarised in Figure F.2.1.

6.5.3.2 Alignment

Once a target is detected, the robot enters the ALIGNING state and rotates towards the opponent using the fuzzy bearing. When a front sensor can see the target, the bearing is taken from the difference between the left and right front readings. If only a side sensor can see it, the combined front and side bearing is used instead. The dead-band holds the previous turn command so the robot does not oscillate about the target direction, and the hold time prevents it from charging at a momentary or spurious reading. The target-lost delay tolerates brief sensor dropouts, so the robot does not abandon a valid target because of a single missed reading.

6.5.4 Attack Algorithm

Once the robot was aligned with the opponent, the attack was divided into four sub-states. In APPROACHING, the robot closes the distance towards the target. In ATTACKING, the robot rams the opponent at full power. In PUSHING, the robot maintains contact and keeps pushing the opponent towards the edge of the dohyo. WOODPECKER is used to break a stalemate where the robot alternates short forward and reverse pulses when the push stalls. The sub-state machine is shown in Figure F.2.1. The transition from APPROACHING to ATTACKING uses hysteresis. The robot enters the attack when the nearest front sensor reads 200 mm or less and only leaves it again when the target moves beyond 250 mm. Without the 50 mm gap, a reading hovering around a single threshold would make the robot switch back and forth between the two states. If the target is lost, the robot does not stay in the attack. When the front sensors lose the target but a side sensor still sees it, the robot returns to ALIGNING. When no sensor sees it, the robot returns to SEARCHING. The following sections describe each sub-state in more detail.

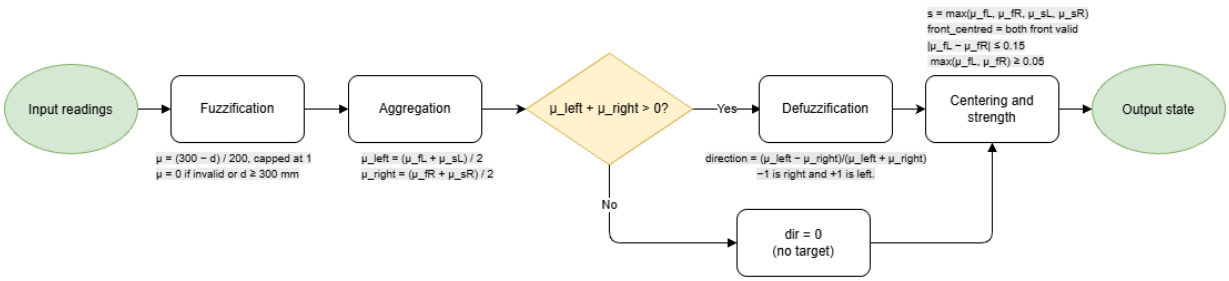


Figure 6.5.2: Fuzzy logic for alignment

6.5.4.1 Approach

In the APPROACHING state, the forward speed was set as a function of the strongest closeness reading s (see Figure 6.5.2).

$$f = 40 + 215\,s \tag{6.5.1}$$

where f is the forward command out of 255 and s ranges from 0 (target at the edge of detection range) to 1 (target at 100 mm or closer). The robot therefore moves slowly when the target is far away and speeds up as it gets closer, reaching full speed as it arrives. Because the closeness is computed from the same ToF readings that steer the robot, a lower speed at long range gives the controller more time per measurement to correct its heading before the target is reached while the increasing speed ensures that the robot hits the opponent with as much momentum

as possible. The robot is steered towards the target using the difference between the front-left and front-right closeness values:

$$u = 150\,(\mu_{fL} - \mu_{fR}) \tag{6.5.2}$$

where a positive u turns the robot counter-clockwise (left). If the target is more to the left, μ_{fL} is larger and the robot turns left and the turn command shrinks to zero as the target becomes centered.

6.5.4.2 Ram and Impact Detection

In the ATTACKING state, the robot rams the opponent at full power (255) with the heading PID bypassed. Once the robot is in contact with an opponent, the heading measured by the IMU is no longer a reliable indication of where the robot should be pointing, because the opponent's push rotates the robot. Any correction the controller made would also take power away from the drive motors and reduce the pushing force. Contact with the opponent was detected using the IMU. The forward acceleration was compared with the value from the previous IMU sample and a change of at least 0.8 g was treated as an impact. When an impact was detected, the robot moved on to the PUSHING state. If no impact was detected within 500 ms of the attack starting, for example because of a glancing contact or a very light opponent, the robot switched to the WOODPECKER state to try to establish contact.

6.5.4.3 Push and Woodpecker

After an impact was detected, the robot entered the PUSHING state and continued to drive at full power. Progress was measured using the front ToF range. As long as the distance to the opponent grew by at least 15 mm from a reference value, the opponent was considered to be moving backwards and the reference was reset to the new range. If no progress was made for 700 ms while the target was still visible, the push was considered to have stalled and the robot switched to the WOODPECKER state. In the WOODPECKER state, the robot repeatedly hit the opponent with short pulses instead of pushing continuously. Each cycle consisted of a 150 ms forward pulse at a command of 255 followed by a 100 ms reverse pulse at 250. These repeated hits help to break the opponent's traction or dislodge its wedge. As soon as the range to the opponent grew by 15 mm again, showing that the opponent was moving away, the robot returned to the PUSHING state.

6.5.5 Back-off Algorithm

6.5.5.1 Edge Detection

The edge was detected with a digital line sensor connected to GPIO13, which reads LOW when the surface under the sensor is white. The detection was split into two layers. The first is a reflex layer: a dedicated edge task polls the sensor every 1 ms and as soon as it sees white it latches an edge stop and commands full reverse directly to the motors, bypassing the control loop. The second is a supervised layer, where the control loop takes over and carries out the recovery manoeuvre described in the next section. Because the reverse command is issued by the edge task and not by the control loop, the worst-case reaction time from the sensor seeing white to the reverse command reaching the motors is 1 ms.

6.5.5.2 Edge Recovery Manoeuvre

After the edge was detected, the robot entered the EDGE_RECOVERY state. It reversed for at least 600 ms and until the sensor had read black for 30 ms, which filters out noisy readings. It then pivoted 180° using the heading PID, and the turn was complete once the heading error was within 3° and the yaw rate below 12°/s for 100 ms. The edge stop was then unlatched and the robot returned to SEARCHING. Two timeouts act as a fail-safe: if the sensor did not read black within 1.5 s of reversing, or the turn took longer than 5 s, the robot switched to the STOPPED state and turned off the motors. These cases indicate that the robot is stuck, for example hanging over the edge or blocked by an opponent, where repeating the manoeuvre could do more harm than stopping.

6.5.6 Telemetry

While the ESP32 can transfer data via a Wi-Fi connection, this module draws up to 250 mA (Random Nerd Tutorials, n.d.). To avoid excessive power consumption, the data gathered per timed run was stored locally via the onboard flash memory and retrieved after the run. At every control cycle (104 Hz) a 32-byte record containing the heading, heading error terms, P, I and D contributions, motor outputs, ToF ranges, forward acceleration and the current state was stored in RAM. Writing to flash during a run stalls the ESP32 and would add jitter to the control loop, so the buffer was only written to a dedicated flash partition once the motors had stopped. A few seconds is allowed to ensure the write completes before the system is turned off. After each run, the partition was read back over USB with esptool (921600 baud) and plotted in Python and MATLAB (Chapter 7).

Chapter 7

Testing and Refinement

This section will discuss the tests and findings related to the performance and behaviour of the sumo robot. Some sections may discuss tuning and further refinements related to these tests, while other sections will summarize the overall performance and conclusions.

7.1 PID Controller Application

As mentioned in Section 6.4, the PID controller was implemented to control the speed of the motors according to the yaw measured by the LSM6DS3 IMU. While preliminary simulations showed that a K_p value of 16 and a K_d value of 0.6 provided the desired response to a step input, the gains were still tuned experimentally.

7.1.1 Experimental Setup and Procedure

The heading controller was evaluated using a straight-line driving test in which the robot was commanded to hold a constant heading of 0°. The duty cycle of the motors was initially set at 200 (out of 255). The test was conducted outside of the dohyo on a flat surface to allow for a longer run time of 6–7 s. During each run, the desired heading, the yaw measured by the LSM6DS3 IMU, the heading error, the individual P, I and D contributions, the motor duty cycles and the forward acceleration were logged after each run. The gains were tuned iteratively. Starting from a proportional-only controller, K_p was increased until the robot held its heading, after which K_d was introduced to damp the oscillations. K_i was kept at zero, in line with the simulation results in Section 6.4. Tuning was considered complete once the heading error remained within the TPM_12 range of 0–10° for the full run.

7.1.2 Experimental Results

Figure 7.1.1 shows the heading without the PID controller, while Figure 7.1.2 shows the heading with the PID controller implemented. The heading error remained within approximately 5° for the entire run, with most of the run within ±2° - which satisfies TPM_12. The ideal parameters were found to be $K_p = 15$, $K_d = 0.6$ and $K_i = 0$. The final value of K_p was slightly lower than the simulated value of 16, since the the oscillations were more pronounced in the real system due to sensor noise and unmodeled dynamics.

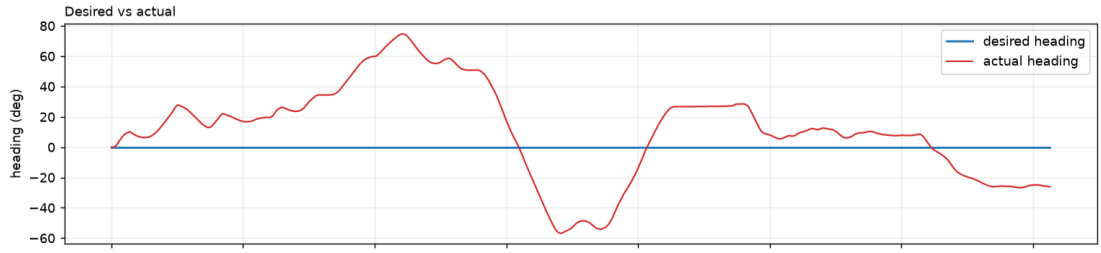


Figure 7.1.1: Heading without PID Controller

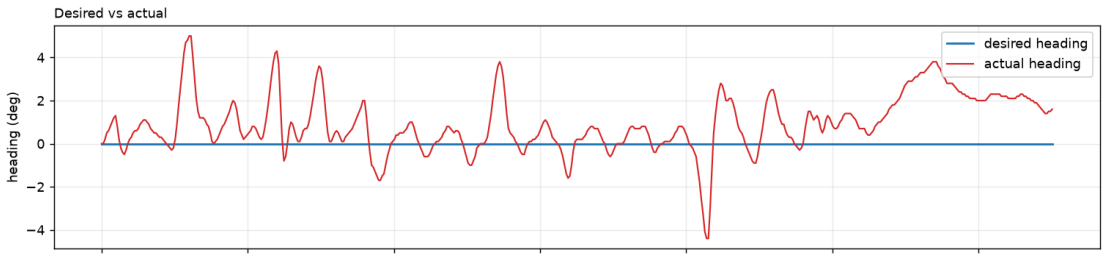


Figure 7.1.2: PID Controller Results

Furthermore, short error spikes of 4–5° coincide with surface irregularities and are corrected within 200 ms. Secondly, the spikes near the end of each run coincide with the stop-button being pressed and the bot picked up. Readings before these spikes were used to evaluate the overall performance of the bot.

7.2 Pivot and Turning Performance

While the previous section focused on the heading control during straight-line motion, this section evaluates the robot's ability to perform in-place pivots at varying speeds.

7.2.1 Experimental Setup and Procedure

With the robot stationary, a step of 90° was applied to the desired heading (see Section C.4). Additionally, the response was tested for smaller angles as needed during the ALIGNING state. Since 90° is larger than most typical alignment corrections, it provides a worst-case scenario for evaluating the pivot performance at varying speeds.

7.2.2 Experimental Results

The same PID controller gains as in the previous section were used for these pivot tests. Figure 7.2.1 shows the response at speeds above 200 duty cycle. Figure 7.2.2 shows the response at speeds of 170 duty cycle (see Section 7.5 for why this duty was of interest). After a dead time of approximately 40 ms, the heading rises

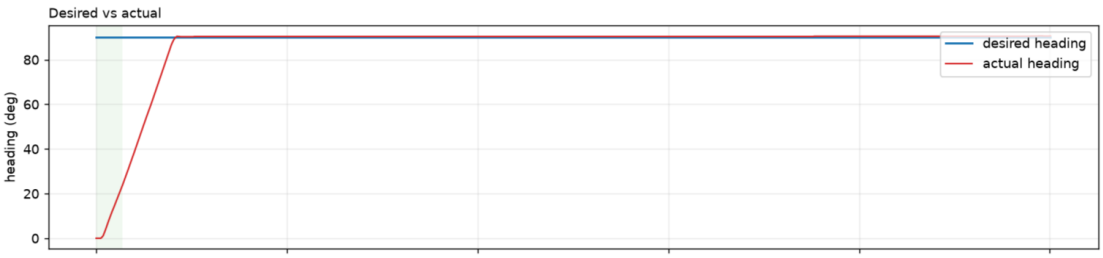


Figure 7.2.1: Response to a 90° step at 170 DC

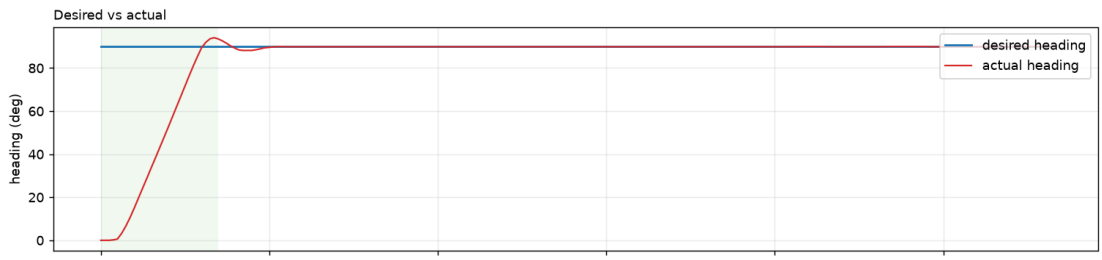


Figure 7.2.2: Response to a 90° step at 250 DC

almost linearly, reaching 90% of the setpoint after approximately 190 ms. The heading peaks at 94° at $t \approx 0.34$ s, an overshoot of 4.5%, followed by a slight undershoot to 88°. The heading settles within ±2% of the setpoint after 470 ms and the steady-state error is zero for the remainder of the run. The results are summarised in Table 7.2.1 and all three transient measures fall within their TPM ranges.

Table 7.2.1: Pivot turn step response results

TPM ID	Measure	Measured	Range	Result
TPM_12	Steady-state error	$\approx 0°$	$\leq 10°$	Pass
TPM_13	Overshoot	4.5%	$\leq 10\%$	Within range (above target)
TPM_14	Settling time	470 ms	≤ 500 ms	Within range (above target)
TPM_15	Rise time	190 ms	≤ 300 ms	Pass

The measured rise time of 190 ms is shorter than the 240 ms predicted by the simulation in Section 6. This difference is expected, because the simulation used roughly estimated motor parameters and did not include the duty limit.

7.3 Maximum Robot Speed

7.3.1 Experimental Setup and Procedure

The robot was placed at the edge of the dohyo and driven straight across it at full duty (255). A video was recorded per run and the time taken to cover the distance was measured from there. This test was repeated 5 times.

7.3.2 Experimental Results

The results are given in Table 7.3.1. The mean speed of 0.404 m/s falls within the TPM_07 range of 0.10–0.77 m/s. Since each run includes the acceleration from rest, this value is a slight underestimate of the true top speed.

Table 7.3.1: Maximum forward speed results at full duty (255)

Run	Distance (m)	Time (s)	Speed (m/s)
1	0.67	1.54	0.435
2	0.67	1.68	0.399
3	0.67	1.72	0.390
4	0.67	1.66	0.404
5	0.67	1.72	0.390
Mean		**1.66**	**0.404**

7.4 Dohyo Edge Detection Response

7.4.1 Experimental Setup and Procedure

The robot was driven towards the edge of the dohyo at full duty (255).Detection was considered successful if the sumo robot stopped before any wheel left the dohyo. The test was first performed with the motors simply stopped on detection and then repeated after the response was refined. Each configuration was tested 3 times.

7.4.2 Experimental Results

When the QRE1113 sensors were tested separately, they reliably detected the transition between the dohyo surface and the white border while meeting TPM_06. However, when the motors were simply stopped, the robot's momentum carried it over the edge of the dohyo. The response was therefore changed to a momentary brake in which full reverse duty is applied as soon as white is detected.



(a) Motors stopped (b) Momentary brake

Figure 7.4.1: Edge detection response with (a) the motors stopped and (b) a momentary brake.

With this change, the robot stopped and remained in the dohyo in every run. The results are summarised in Table 7.4.1.

Table 7.4.1: Edge detection response

Response	Runs	Edge detected	Stayed in dohyo
Motors stopped	3	3/3	0/3
Full reverse (momentary brake)	3	3/3	3/3

7.5 Opponent Detection

7.5.1 Experimental Setup and Procedure

The maximum search spin speed is limited by the time the ToF sensors need to take a measurement. If the robot spins too fast, the opponent passes through the sensor's field of view before enough readings are taken to confirm it. Each VL53L0X has a field of view of approximately 25° and a measurement time of 35 ms. To confirm a target, the opponent must remain in view for 2 consecutive readings, which gives a theoretical maximum spin rate of

$$\omega_{max} = \frac{\theta_{FOV}}{n\, t_{meas}} = \frac{25^\circ}{2 \times 35 \text{ ms}} \approx 357^\circ/\text{s} \tag{7.5.1}$$

To verify this, an opponent was placed 300 mm from the robot, and the robot was spun in place in the SEARCHING state at different duty values. Each duty was tested 5 times, and a test was counted as successful if the robot detected the opponent and stopped spinning on its first pass.

7.5.2 Experimental Results

The results are given in Table 7.5.1. Detection was reliable up to a duty of 170 (about 340°/s), above which the opponent was missed. A search duty of 170 was therefore selected. At this speed, the worst-case detection time is 70 ms, within the TPM_02 range of 0–100 ms.

Table 7.5.1: Opponent detection reliability at different search spin speeds

Duty	Spin rate (°/s)	Detected
130	260	5/5
150	300	5/5
170	340	5/5
200	400	2/5
255	510	0/5

7.6 Pushing Force

7.6.1 Experimental Setup and Procedure

Weights of 100, 200, 300, 400 and 500 g were placed in front of the robot. The robot was then driven forward at full duty (255). A test was successful if the robot pushed the weight continuously across the dohyo without stalling. 500 g was chosen as the upper limit, since it is the maximum robot mass permitted by the competition rules.

7.6.2 Experimental Results

The results are given in Table 7.6.1. The robot pushed every load, including the 500 g load, without stalling, showing that it can push an opponent of the maximum allowed mass.

Table 7.6.1: Pushing test results at full duty (255)

Mass (g)	Pushed successfully
100	Yes
200	Yes
300	Yes
400	Yes
500	Yes

7.7 Additional Tests

While the six main tests were described, additional tests were conducted throughout the project.

Power-Supply Tests The regulators were tested under different voltage inputs set by a DC-power supply. This is where the original buck-converters failed, but the linear regulators proved reliable.

VL53L0x Sensor Test The VL53L0x sensors were tested individually to verify their range and accuracy. The test involved placing objects at known distances and recording the sensor readings to ensure they matched the expected values. Some of the sensors needed to be calibrated after this test, but otherwise they operated reliably.

QRE1113 Sensor Test The sensors were tested under varying light conditions to ensure they could reliably detect the edge of the dohyo. The sensors passed every test excluding illuminance below 10 lux (see Section C.2), confirming that the sensors could operate relaibly - especially within the enclosed frame.

Friction Tests The friction of the wheels was tested on the dohyo surface to determine the slip angle and the corresponding static friction coefficient. The results are summarized in Section C.3. This was mainly done to justify the choice between polyurethan 40 A and 30 A wheels.

Heavier Object While the main goal was to push 500 g, in real competitions, that 500 g would be pushing back. Similar to the test in Section 7.6, a heavier object was placed infront of the sumo-robot during autonomous runs. During these runs, the sumo robot would not always be able to push the heavier object. In these cases, the woodpecker-method would allow the robot to gradually chip away at the opponent's position, eventually gaining an advantage despite the initial resistance.

Opponent Detection Tuning While the detection range was originally designed for the span of the arena (TPM_09), this needed to be decreased to 300 mm to avoid false positives from objects outside of the arena.

Competition Rules While this was not an iterative test, both the final weight and size were measured (see Appendix A). Both the weight and size complied with the competition rules, meeting TPM_03 and TPM_04.

7.8 Autonomous Matches

During the last stages of testing, the sumorobot was finally tested in autonomous matches against other robots. The matches were conducted on the standard dohyo with each robot passing the initial rule-adherence checks.
The implemented sensor array and control logic allowed for much faster 'lock-ons' and adjustments compared to some of the other robots. Furthermore, while not every match was won, the frame successfully protected the electrical components from impacts and collisions throughout the matches. In the cases where the matches were lost, it was due to motor limitations. Despite this, the robot demonstrated reliable autonomous behavior and effective use of its sensors throughout the matches - winning a total of 4 out of 6 practice matches.

Chapter 8

Conclusions and Future Work

While this project achieved its goal in setting a stable foundation - there is still much that can be improved. After several autonomous runs against opponents, it was observed that the sumobot's response was rapid and accurate compared to the other bots. However, bots using stronger motors were still able to win more consistently regardless of their slower responses. Ultimately, replacing the N20 motors with brushless motors would likely improve the robot's overall performance in competitions.

Furthermore, now that the sensor-array and other electronics were successfully integrated and tested, future work could focus on implementing the PCB design to streamline the wiring, improve reliability and reduce the overall footprint of the robot.

Lastly, while the base finite-state control for more adaptive combat was successful, this could always be improved further as more experience is gained regarding opponents and match strategies. There will always be a new opponent, an unseen strategy or an unexpected situation. Without the time-constraint, this project could never truly be done.

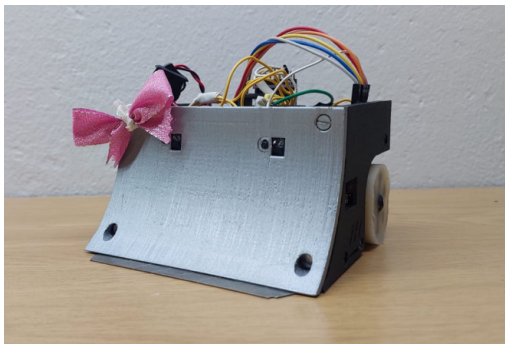


Figure 8.0.1: The Final Built Sumo Robot ('Jeremy')

Appendix A

Final Built Sumo Robot

A.1 Competition Restraints: Size and Weight

The following figures display that the final built sumo robot meets the competition restraints in terms of weight and size - meeting TPM_03 and TPM_04 (FR_07).



Figure A.1.1: Final built sumo robot weight.

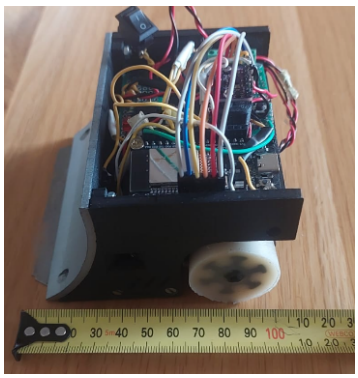


Figure A.1.2: Final built sumo robot size.

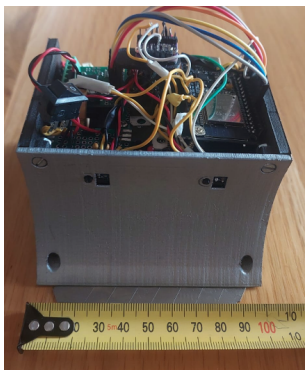


Figure A.1.3: Final built sumo robot height.

Appendix B

Evaluation Criterion

B.1 Concept Design Evaluation

Table B.1.1: Evaluation criteria and their justification

Criterion	**Reason**
Manoeuvrability	The robot must move and turn accurately within the dohyo while supporting the goal to out-manoeuvre the opponent (FR_03).
Traction	The robot must push the opponent out of the dohyo (FR_04). More traction helps maintain control and apply force effectively.
Complexity	Fewer motors reduce the mass and wiring, which helps the robot stay within the weight limit (FR_07) and makes it simpler to build.
Integration	All components must fit within the 100×100 mm footprint (FR_07).
Range and coverage	The robot must detect the opponent (FR_01) anywhere in the dohyo, up to 770 mm and across a wide enough field of view not to miss an opponent approaching from the side.
Response time	The measurement time of each sensor type limits how soon the robot can respond.
Robustness	Detection must stay reliable despite ambient light, the opponent's surface reflectivity and interference between sensors (FR_01) since the opponent's shape and material are unknown.
Integration	The sensors and wiring must fit within the 100×100 mm footprint and 500 g mass limit (FR_07).

Continued on next page

Table B.1.1 continued from previous page

Criterion	**Reason**
Distance output	A measured distance, rather than only a detected/not-detected signal, lets the controller judge how far away the opponent is and decide how to respond (FR_01, FR_05).
Force redirection	The robot must push the opponent out of the dohyo (FR_04) without being stalled or lifted. A wedge that deflects the opponent's force, rather than absorbing it head-on, keeps the robot moving and supports a flanking strategy.
Engagement	How well the wedge gets under, beside or around the opponent determines whether the robot can apply its pushing force (FR_04).
Sensor integration	The wedge shape decides where the sensors can be mounted. The sensors need a clear field of view to detect the opponent (FR_01).
Internal space	The electronics, battery and motors must fit in the chassis within the 100×100 mm footprint and 500 g mass limit (FR_07).
Manufacturability	The wedge must be made with the facilities available to the project (SR8). A simpler shape is quicker to make and easier to repeat if it breaks.
Heading measurement	The robot must keep a straight line and turn by a controlled amount (FR_03).
Wheel speed measurement	Speed feedback lets the controller drive both wheels at the commanded speed (FR_03), which keeps the robot moving straight.
Integration	Sensors, mounts and wiring must fit within the 100×100 mm footprint and 500 g mass limit (FR_07). Wheel encoders need extra parts at each wheel, while a single board does not.
Robustness (slip, light, impacts)	The measurement must stay accurate when the wheels slip, under changing light and during impacts and the sensor must survive the match (FR_03, FR_11).
Interface (pins)	A sensor that uses fewer pins leaves pins free for the other peripherals.

B.2 Refined Design Evaluation

Table B.2.1: Comparison of opponent detection technologies

Parameter	**Ultrasonic HC-SR04**[a]	**IR TCRT5000**[b]	**ToF VL53L0X**[c]
Range	20 mm–4 m	0.2 mm – 80 cm	30 mm – 2 m
Field of view	15°	24–35°	25°
Surface reflectivity sensitivity	Low	High	Moderate
Ambient light sensitivity	Low	High	Low
Typical measurement time	23 ms	10–15 μs	20–33 ms

[a] (ELECFreaks, n.d.; ArcBotics, n.d.)
[b] (AM Electronics, n.d.; Electrobes, 2026)
[c] (STMicroelectronics, 2024)

Table B.2.2: Qualitative comparison of PLA, PETG and TPU filaments

Property	**PLA**	**PETG**	**TPU**
Stiffness	High, rigid	Medium-high	Low, flexible
Strength	Good (brittle)	Good (tough)	Moderate (elastic)
Impact resistance	Low	High	Very high
Flexibility	Very low	Low-moderate	Very high
Heat resistance	Low (∼60 °C)	Moderate (∼80 °C)	Moderate
Cost	Low	Low-moderate	Moderate-high

Table B.2.3: Comparison of wheel materials

Material	**Shore A**	**Density (g/cm^3)**	**Friction coefficient**
Polyurethane[a]	25–90	1.24–1.27	0.5–1.2
Silicone rubber[b]	25–80	1.21	0.4–0.8
SEBS[c]	10–80	0.87–0.93	0.4–1.0

[a] (Zenith Rubber Europe, 2021)
[b] (Revchem Composites, 2022)
[c] (Mexpolimeros, 2025)

Appendix C

Testing

C.1 VL53L0x Testing

The VL53L0x sensors were tested to measure set distances at various angles. The distances were confirmed using vernier calipers or measuring tape. A protractor was used for the angle. For more test setups, results and videos - see https://github.com/chanelleblan/SumobotSkripsie.git.

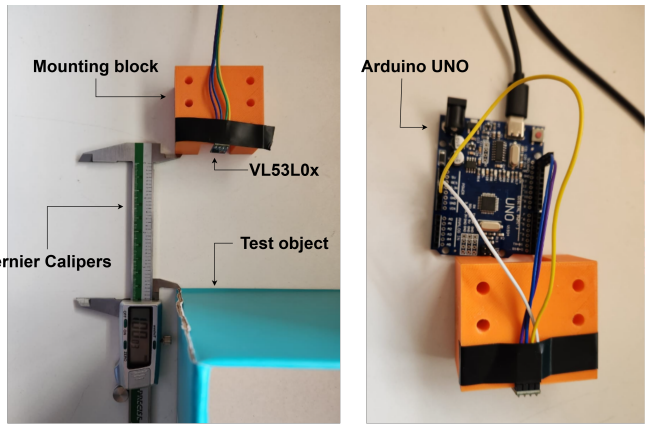


Figure C.1.1: VL53L0x Preliminary Testing Setup

The results are summarized in Table C.1.1.

Table C.1.1: VL53L0X measured distance (cm) versus actual distance and incidence angle

	Actual distance (cm)											
Angle (°)	2	3	4	5	10	15	20	25	50	60	70	80
0	2.1	3.1	4.2	5.1	10.2	15.1	20.3	25.2	50.6	60.5	70.7	80.4
5	2.2	3.0	4.1	5.2	10.3	15.3	20.5	25.4	50.8	60.7	71.0	80.8
10	2.3	3.2	4.2	5.3	10.5	15.6	20.8	25.8	51.5	61.8	72.0	81.6
20	2.4	3.3	4.5	5.7	11.2	16.8	18.4	27.6	43.8	56.5	NaN	NaN
30	2.7	3.8	5.2	6.5	12.8	18.9	29.4	21.7	NaN	NaN	NaN	NaN
45	10.1	88.5	77.3	60.5	NaN	NaN	88.3	NaN	NaN	NaN	NaN	NaN
60	77.5	30.5	NaN	74.3	75.1	NaN	NaN	NaN	NaN	NaN	NaN	NaN

C.2 QRE1113 Sensor Testing

The lux level was measured using an app - 'Light Meter'. 200 lux was measured as the average lighting of the venue, 25 was measured within the enclosed frame and below 10 lux was measured in a completely enclosed and darkened setting (varied around 2–10 lux). While these lux levels might not be completely accurate, they were used as a relative measure.

Table C.2.1: QRE1113 edge detection under varying light conditions

Illuminance (lux)	**White detected**	**Black detected**
< 10	No	Yes
25	Yes	Yes
50	Yes	Yes
100	Yes	Yes
150	Yes	Yes
200	Yes	Yes

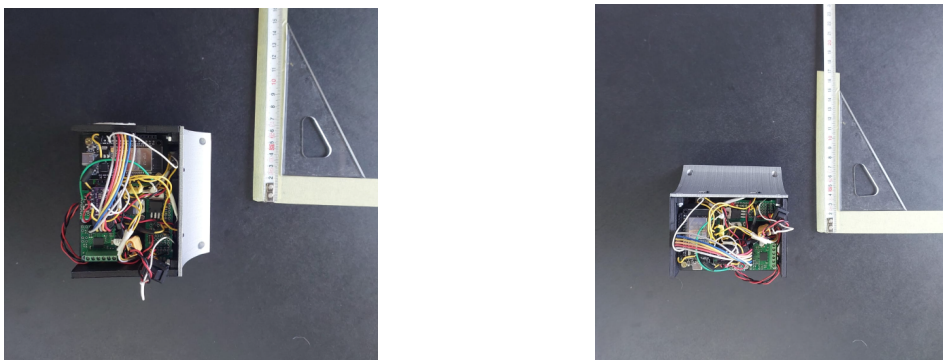
C.3 Friction Tests

Table C.3.1: Incline angle at which each wheel material slipped

Wheel material	**Slip angle (°)**	$\mu_s = \tan\theta$
SEBS	15–20	0.27–0.36
Polyurethane, Shore 40 A	64	2.05
Polyurethane, Shore 30 A	68	2.48

C.4 Turning Tests

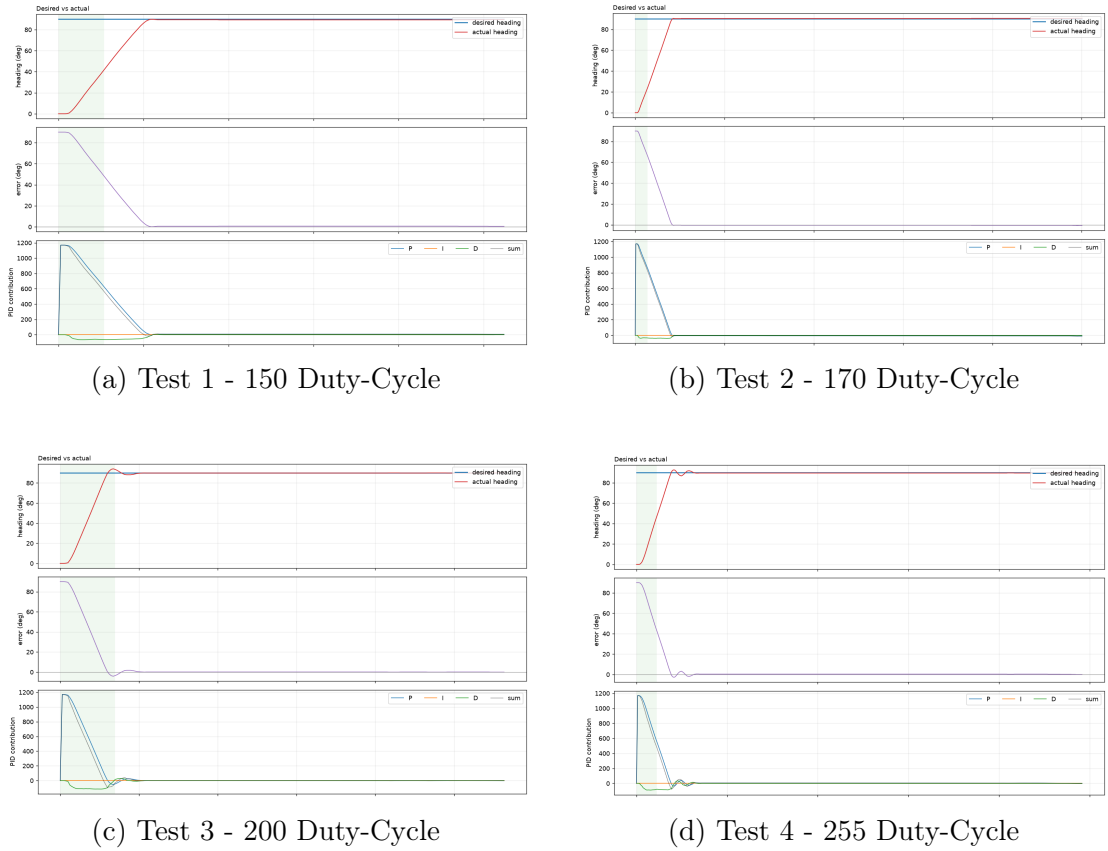
The following figures show the setup for the turning tests. The markers on the arena were there for visual reference and 'sanity checks'.



(a) Before 90° turn. (b) After 90° turn.

Figure C.4.1: Turning Test Setup

The graphs below show the results of the turning tests, indicating the robot's heading before and after executing a 90° turn.



(a) Test 1 - 150 Duty-Cycle

(b) Test 2 - 170 Duty-Cycle

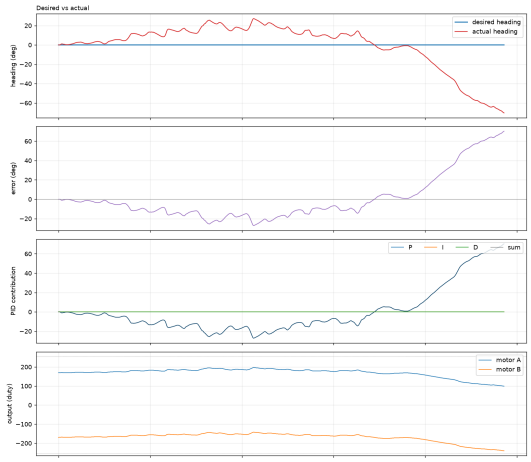
(c) Test 3 - 200 Duty-Cycle

(d) Test 4 - 255 Duty-Cycle

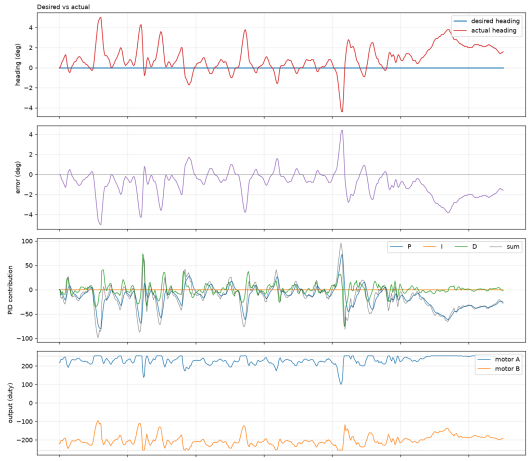
Figure C.4.2: Turning Test Telemetry Results

C.5 Heading Tests

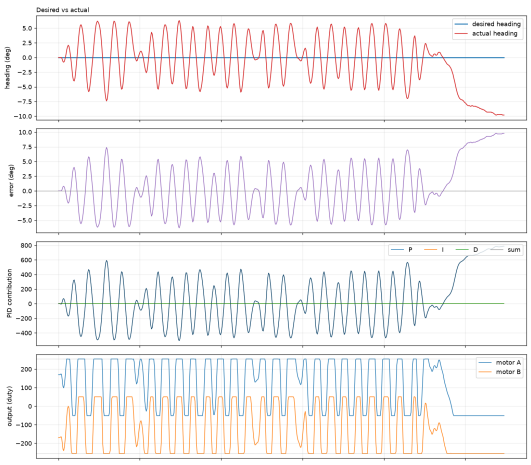
The following figures display some of the tests regarding the robot's heading control under different PID settings. In total, twenty tests were conducted to evaluate the performance.



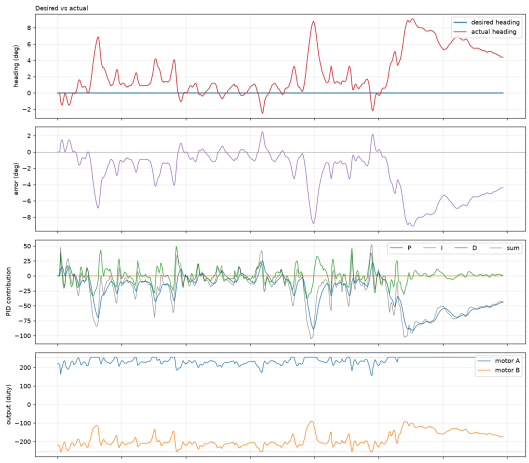
(a) Test 1 - P = 1, I = 0, D = 0



(b) Test 2 - P = 15, I = 0, D = 0.6



(c) Test 3 - P = 80, I = 0, D = 0



(d) Test 4 - P = 15, I = 0, D = 0.5

Figure C.5.1: Heading Test Telemetry Results

Appendix D

Techno-Economic Analysis

D.1 Project Schedule

While planning the project, an initial schedule was created to outline the key milestones and deadlines.

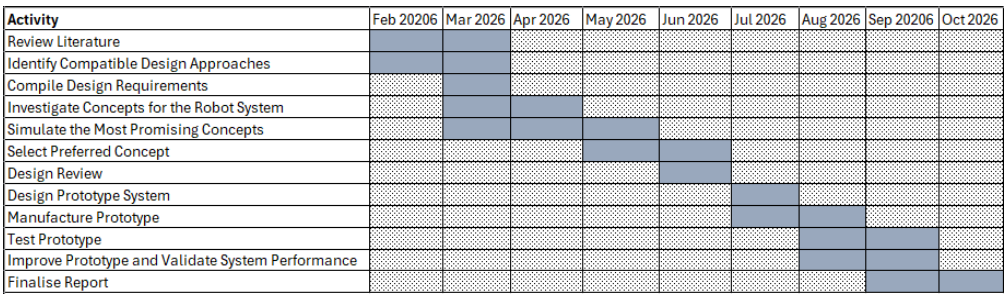


Figure D.1.1: Original Planned Project Schedule.

Throughout the progress of the project, the schedule was adjusted as it became clear that some activities had to be post-poned or completed in parallel with later activities. The project remained on schedule after the revised plan was implemented.

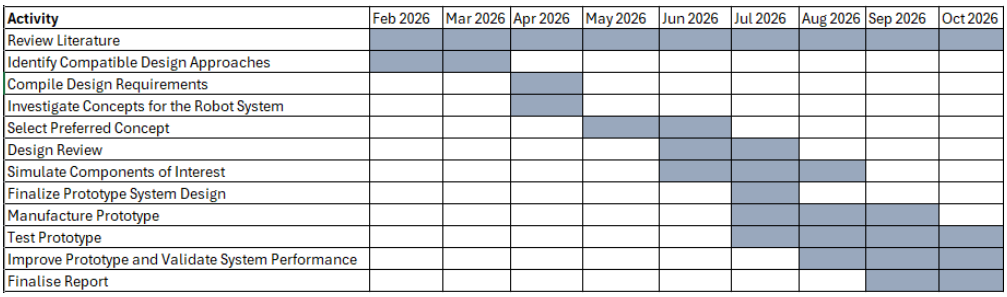


Figure D.1.2: Revised Project Schedule.

D.2 Budget

The proposed budget for the project is outlined in Table D.2.1, showing the theoretical cost of engineering and workshop time. The overall cost is estimated to be R275 955.

Table D.2.1: Theoretical cost of engineering and workshop time

Activity	Engineering Time		Capital Costs	Running Costs	Facility Use	MMW Labour		MMW Material
	hr	R	R	R	R	hr	R	R
Review literature	25	12 500	0	0	0	0	0	0
Identify compatible design approaches	30	15 000	0	0	0	0	0	0
Compile design requirements	20	10 000	0	0	0	0	0	0
Investigate concepts for the robot system	80	40 000	0	250	0	0	0	0
Simulate the most promising concepts	70	35 000	0	500	0	0	0	0
Select preferred concept	10	5 000	0	0	0	0	0	0
Design review	5	2 500	0	0	0	0	0	0
Design prototype system	80	40 000	0	500	0	0	0	0
Manufacture prototype	20	10 000	0	1 000	0	25	7 500	500
Test prototype	30	15 000	500	200	200	10	3 000	0
Improve prototype and validate system performance	50	25 000	0	1 000	200	0	0	0
Finalise report	100	50 000	0	50	0	0	0	0
TOTAL	**520**	**260 000**	**500**	**3 500**	**400**	**35**	**10 500**	**500**

Table D.2.2: Actual project costs.

Activity	**Engineering Time**		**Components**	**Running Costs**	**Facility Use**	**MMW Labour**		**MMW Material**
	h	**R**	**R**	**R**	**R**	**h**	**R**	**R**
Review literature	30	15 000	–	–	–	–	–	–
Identify compatible design approaches	30	15 000	–	–	–	–	–	–
Compile design requirements	25	12 500	–	–	–	–	–	–
Investigate concepts for the robot system	80	40 000	–	250	–	–	–	–
Simulate the most promising concepts	60	30 000	–	500	–	–	–	–
Select preferred concept	10	5 000	–	–	–	–	–	–
Design review	5	2 500	–	–	–	–	–	–
Design prototype system	90	45 000	–	500	–	–	–	–
Manufacture prototype	40	20 000	3 970.17	1 000	–	25	7 500	1 500
Test prototype	50	25 000	–	200	200	10	3 000	–
Improve prototype and validate system performance	80	40 000	–	1 000	200	–	–	–
Finalise report	100	50 000	–	50	–	–	–	–
TOTAL	**600**	**300 000**	**3 970.17**	**3 500**	**400**	**35**	**10 500**	**1500**

The final engineering time cost is R300 000 where the proposed amount was R275 400. Furthermore, components cost a total of R4 970.17 including the cost of PETG to manufacture parts. While the component and manufacturing costs are under the R6 000 budget, the overall project cost exceeds the initial proposed budget, reaching R320 505.17.

D.3 Technical Impact

The project demonstrates a complete autonomous mini-sumo robot that fits within the competition limits of 100 × 100 mm and 500 g. It combines time-of-flight opponent detection, infrared edge detection and an IMU-based heading controller within a state machine that dodges, searches, aligns, attacks and recovers from the dohyo edge. The final robot provides a solid foundation to enter sumo-robot competitions and the findings can be used to further improve the current system and guide the development of future mini-sumo robots.

D.4 Return on Investment

No financial return was measured since the robot is a university project and not a sales product. The return on this investment is a working, tested robot together with reusable CAD models, electronic designs, simulation files, embedded software and measured data. These can shorten the development time of later mini-sumo or mobile-robot projects.

D.5 Potential for Commercialisation

The robot is a prototype and not a market-ready product. Mini-sumo robots are mainly built for competitions. Commercial kits already exist in that market, so a custom 3D-printed robot with hand-assembled electronics is unlikely to compete on cost or on performance. Its potential is mainly as an educational platform for sensor integration, feedback control and embedded software using the documented design and test results. Production would need integrated circuit boards, repeatable and cheaper manufacturing, stronger motors (see Chapter 8) and further durability and safety testing (particularly of the LiPo battery).

D.6 Responsible Use of Resources and End-of-Life Strategy

The robot is modular and uses standard fasteners, replaceable 3D-printed parts and off-the-shelf components so a damaged part can be replaced individually without scrapping the whole assembly.
At end of life, the LiPo battery should be removed first and inspected separately. A serviceable pack can be reused in compatible equipment. A damaged, swollen or exhausted pack must not go into general waste or ordinary recycling bins. Its terminals should be protected against short circuits, and it should be handed to an approved lithium-battery collection or recycling service.
Once the battery is removed, the robot can be dismantled with standard tools. Reusable parts should be recovered first: the N20 motors, the motor driver, sensors, the IMU, the microcontroller board, wiring and fasteners. Faulty circuit boards and electronic components should go to electronic-waste recycling. The printed parts (PETG) should be separated from metal fasteners. The polyurethane wheels cannot be remelted, so they should be disposed of responsibly if they cannot be reused. This approach aims to keep useful components in use and reduce electronic, battery and polymer waste.

Appendix E

Calculations and Simulations



Figure E.0.1: Motor-related Calculations.

Created using a free version of SMath Studio

Battery Voltage

$V_{max} := 12,6\ V$

$V_{min} := 9,3\ V \quad V_{nominal} := 11,1\ V$

Current and Discharge $750 \cdot mAh \quad 45 \cdot C$ from listing

$I_{battery} := \left(750 \cdot (10)^{-3}\right) \cdot 45\ A \qquad I_{battery} = 33,75\ A$

$I_{motorstall} := 500\ mA$ Lower end estimate

$I_{MCU} := 25\ mA \quad (max) \qquad I_{IR} := 40\ mA$

$I_{TOF} := 40\ mA$ Startup and shutdown peak $\qquad I_{IMU} := 3,6\ mA$

$I_{motordriver} := 3\ A$

$I_{system,peak} := 2 \cdot I_{motorstall} + I_{MCU} + 5 \cdot I_{TOF} + 3 \cdot I_{IR} + I_{IMU} + I_{motordriver}$

$I_{system,peak} = 4,3486\ A$

$SF_{current} := \frac{I_{battery}}{I_{system,peak}} \qquad SF_{current} = 7,7611$

Normal operation and run time est

$t_{run} := \frac{C_{3s}}{I_{max}} \quad C_{3s} := 750\ mA\ hr \quad I_{max} := I_{system,peak}$

$t_{run} = 620,8895\ s$

Figure E.0.2: Battery-related Calculations.

Thermal Calculations (heatsink)

$I_{5Vline} := I_{MCU} + 2 \cdot I_{IR} + I_{IMU} + 4 \cdot I_{TOF}$

Voltage measured at 11.49V after diode from LTSpice simulation

$I_{5Vline} = 0,2686\ A \qquad V_{afterdiode} := 11,49\ V$

$I_q := 0,005\ A$ Regulator datasheet (5 V)

$V_{out} := 5\ V$

$P_D := \left(V_{afterdiode} - V_{out}\right) \cdot I_{5Vline} + V_{afterdiode} \cdot I_q$

$R_{CS} := 0,5\ \frac{°C}{W} \qquad R_{SA} := 50\ \frac{°C}{W} \qquad P_D = 1,8007\ W$

$R_{JC} := 5\ \frac{°C}{W}$

$T_{J1} := P_D \cdot \left(R_{JC} + R_{CS} + R_{SA}\right) \qquad T_{J1} = 99,9369\ °C$

Final junction temperature is around 110 deg at maximum operating conditions (TJ1+TA). 5 V regulator's maximum rating is 125

$T_{SF} := \frac{99,93}{125} \qquad T_{SF} = 0,7994$

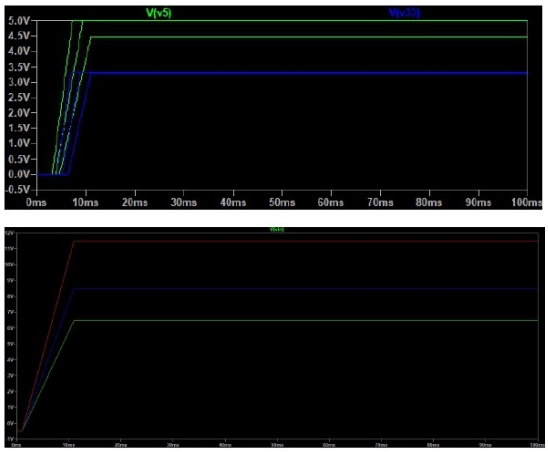


Figure E.0.3: Heatsink Calculations.

Created using a free version of SMath Studio

Real friction coefficient

$\theta_{shore30} := 68\ deg \quad m_{shore} := 10\ g$

$\theta_{shore40} := 64\ deg \quad m_{sebs} := 6\ g$

$\theta_{sebs} := 15\ deg$

$f_{s30} := m_{shore} \cdot g \cdot \sin\left(\theta_{shore30}\right)$

$f_{s40} := m_{shore} \cdot g \cdot \sin\left(\theta_{shore40}\right)$

$f_{ssebs} := m_{sebs} \cdot g \cdot \sin\left(\theta_{sebs}\right)$

$f_{s30} = 0,091\ N$

$f_{s40} = 0,0882\ N$

$f_{ssebs} = 0,0152\ N$

Maximum static friction

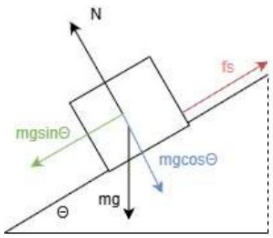
$\mu_{s30} := \frac{f_{s30}}{N_{30}} \quad \mu_{s30} = 2,4751$

$\mu_{s40} := \frac{f_{s40}}{N_{40}} \quad \mu_{s40} = 2,0503$

$\mu_{ssebs} := \frac{f_{ssebs}}{N_{sebs}} \quad \mu_{ssebs} = 0,2679$







$N_{30} := m_{shore} \cdot g \cdot \cos\left\{\theta_{shore30}\right\}$

$N_{40} := m_{shore} \cdot g \cdot \cos\left\{\theta_{shore40}\right\}$

$N_{sebs} := m_{sebs} \cdot g \cdot \cos\left(\theta_{sebs}\right)$

$b := \begin{bmatrix} \mu_{s30} \\ \mu_{ssebs} \\ \mu_{s40} \end{bmatrix}$

$\mu_{preferred} := \max(b)$

$\mu_{preferred} = 2,4751$

Figure E.0.4: Friction Test Results for Wheel Materials.

Appendix F

Software Flow Diagrams

F.1 Task Structure Overview

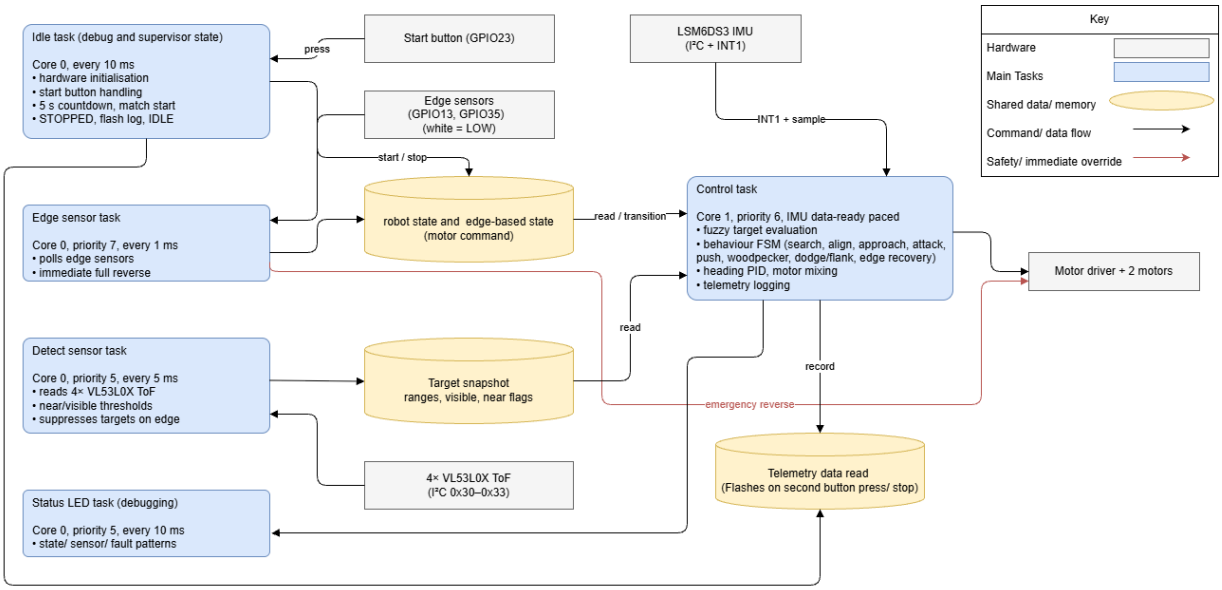


Figure F.1.1: Task Structure Overview

F.2 State Machine Overview

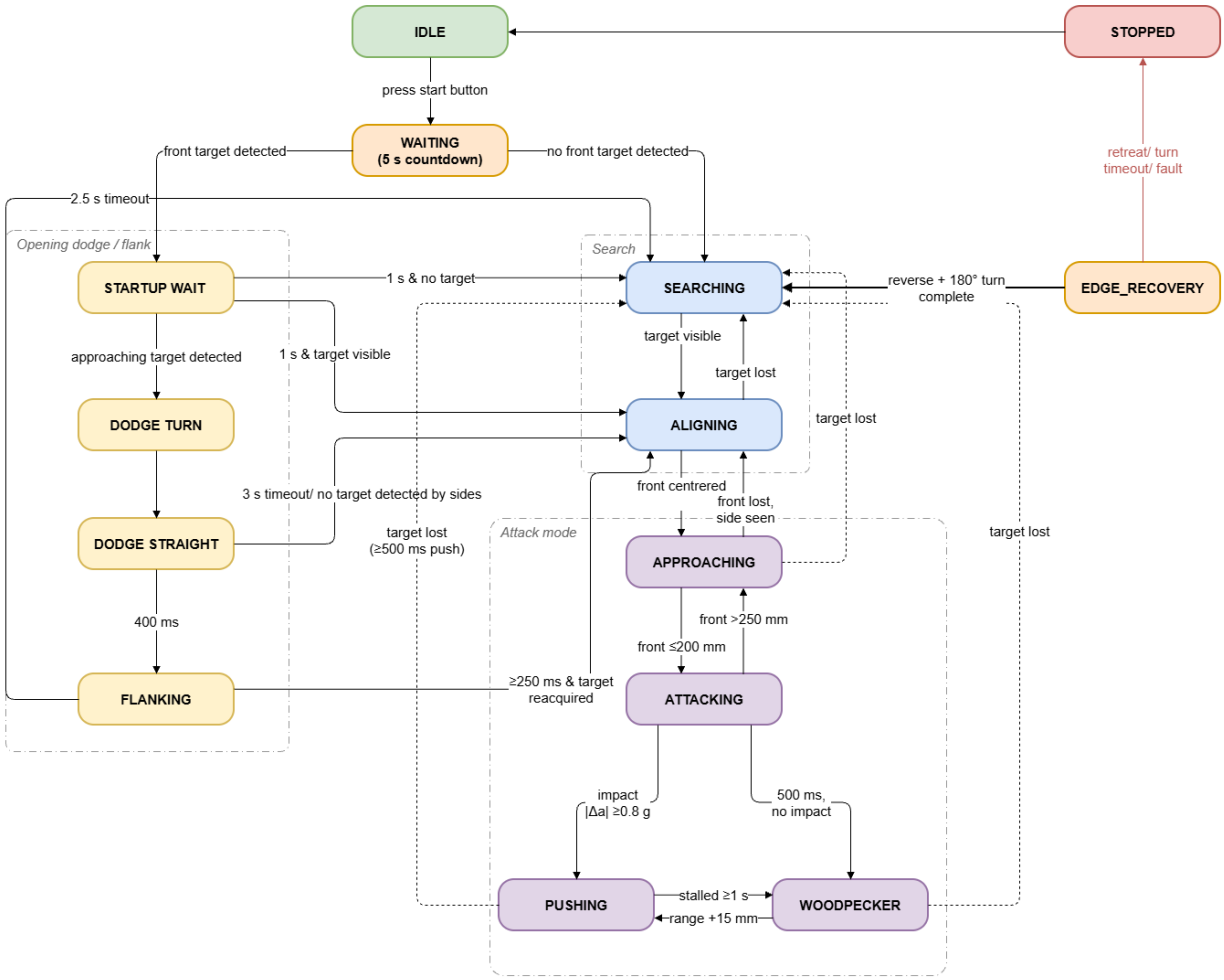


Figure F.2.1: State Machine Overview

F.3 Initialization flow

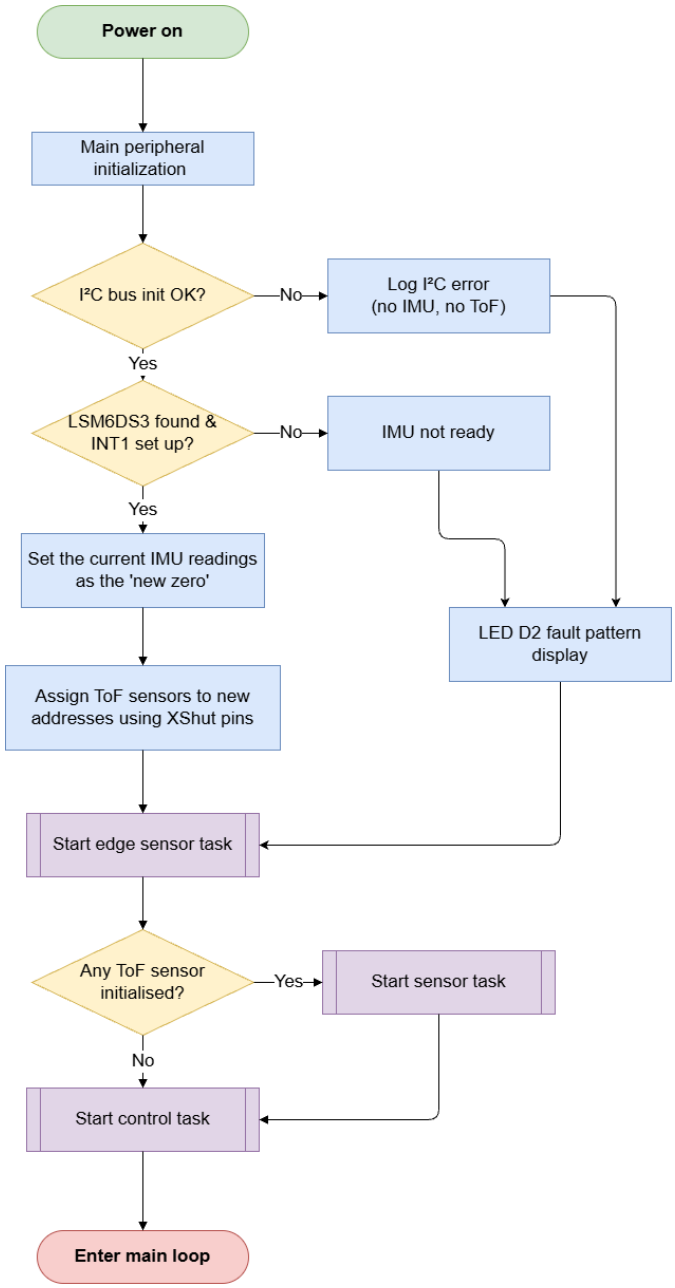


Figure F.3.1: Initialization Flow Diagram

Appendix G

Electrical Schematic

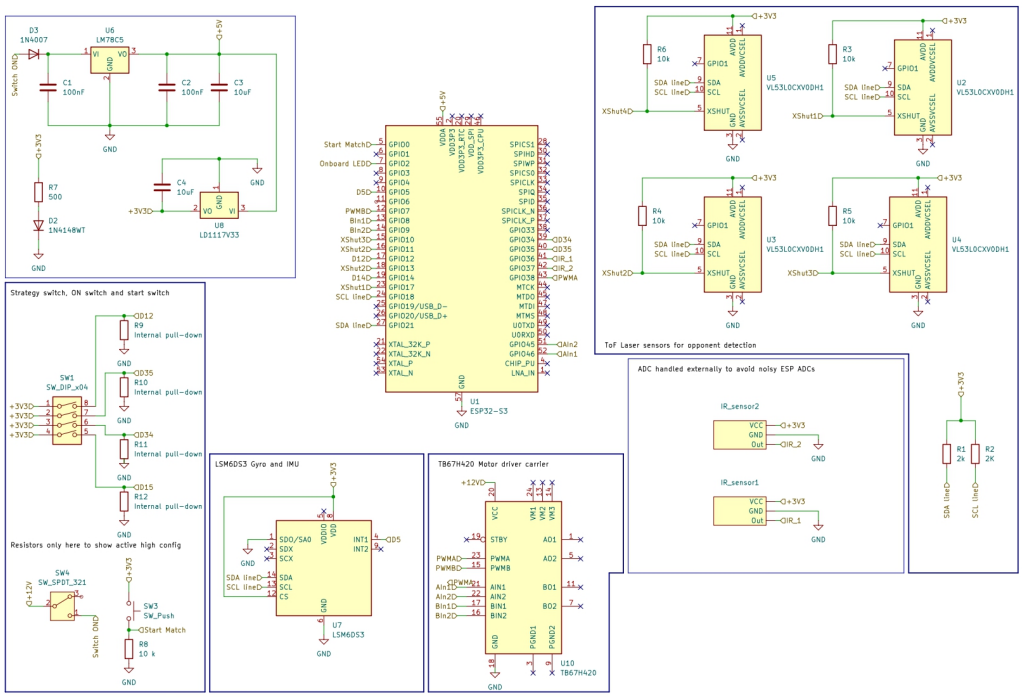


Figure G.0.1: Electrical Schematic

Appendix H

Safety Report and Risk Assessment

Stellenbosch
UNIVERSITY
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UNIVERSITEIT

Autonomous Sumobot Safety Report

Autonomous Sumobot Practical Work and Arena Testing

Date:	16/09/2026
Student & SU nr:	Chanelle Blanche 25929291
Student contact nr:	072 123 0339
Supervisor:	Mr Wayne Swart
Lab engineer:	Mr Kevin Neaves
Head of safety:	Mr Cobus Zietsman

Emergency Contacts:

Contact:	Room nr.	Work nr.	Cell nr.
Kevin Neaves	M3001	-	
Mr C. Zietsman	M212	021 808 4275	-
Campus Security	-	021 808 2333	WhatsApp: 082 808 233
Fire Brigade	-	021 808 8888	-
Ambulance	-	021 883 3444	-

Signatures:

Student:
(C Blanche)

Supervisor:
(W Swart)

Lab Engineer:
(K Neaves)

Head of Safety:
(C Zietsman)

KMN 1/10/2026

Pressure vessels or pipes (check relevant box):
☒ No pressure vessels or pipes with pressure in excess of 50kPa are involved in this project.
☐ Pressure vessels or pipes in excess of 50kPa are involved – additional signature and report required (refer to Safety Report Guidelines on SUNLearn).

Hot work / working at heights / confined entry / excavation (check relevant box):
☒ No hot work / working heights / confined entry / excavation work involved in this project.
☐ Hot work / working heights / confined entry / excavation work (underline relevant work type(s)) involved in this project – additional signature and report required (refer to Safety Report Guidelines on SUNLearn).

Warning Symbols

Applicable warning signs:



(a) Electrical hazard (b) Flammable substance (c) General warning

Figure 1: Applicable warning/hazard symbols

General Laboratory Safety

The following general laboratory safety instructions are applicable:

- No afterhours testing may be performed without the necessary permissions.
- Full supervised training is required before testing may be undertaken. Permission to proceed with unsupervised testing should be signed off by the lab engineer.
- Closed shoes must be worn at all times.
- Emergency equipment must be located and easily accessible.
- Students may not work alone in the laboratory.
- Emergency exits must be known. The nearest exits applicable to the setup are provided in Appendix A of this document.
- Loose clothing may not be worn. Loose hair must be tied up.
- Good housekeeping practices should be maintained during testing. The lab should be completely clean, including all equipment stored away, after testing. Refer to the General Housekeeping section for particulars regarding practices to be followed for this specific setup.
- No food or drink is permitted in the laboratory.
- Safety report must be visible and accessible during testing.
- No equipment or test may be left unattended.

Anticipated Interactions with other Laboratory Users

The sumobot will be tested in a laboratory with a prepared arena (room 224). Other sumobot-students may be conducting similar experiments relating to the arena setup. Alongside the arena, a charging station for batteries has been set up. Apart from the sumobot-students, other bot-students may be present to use these chargers as well. Furthermore, lab staff may be present.

General Housekeeping

The workspace will be kept clean and free of loose wires, tools and other objects that could interfere with the tests previously mentioned as well as other students using the area. The sumobot, battery and other equipment will be stored safely when not in use. Tape and other temporary test markers will be removed when testing is complete.

Fire Safety

The main fire-related hazard is the LiPo battery which can become damaged, overheat or catch fire if incorrectly charged, short-circuited or physically damaged. The battery will be inspected before, during and after use. Cell voltages will be monitored and a fireproof bag will be used to store the battery during transportation and charging. The nearest fire extinguishers have been identified in Appendix A as well.

Activity Based Risk Assessment

Activity	Risk	Risk Type (P/E)	Mitigating Steps	Classification of Risk Severity
Powering the system with a LiPo battery	Flammable/ a fire hazard	E	• Inspect battery before use for physical signs of damage (swelling etc). • Use an appropriate LiPo charger. • Disconnect battery when not in use. • Inspect individual cell voltages after every use (above 3.8 V per cell.)	Medium
Operating the sumobot	Robot moving unexpectedly	P	• Stay away from the arena once the robot has started. • A time-out mechanism has been implemented to stop the robot after nothing has been detected for 10 s – allowing it to be retrieved safely.	Low
Collision between the sumobot and test objects	Damage to robot components or loose parts	E	• Keep the testing area/ arena clear. • Inspect wiring and other hardware components after each test.	Low

Activity	Risk	Risk Type (P/E)	Mitigating Steps	Classification of Risk Severity
Electrical connections and wiring	Short circuit, overheating or component damage	E	• Check wiring before powering the system • Use a multimeter and check for shorts between the main power rails. • Disconnect the battery when changing wiring.	Medium
Testing at high speed	Robot may leave the intended testing arena	P	• Test within the arena and keep the surrounding area clear.	Low
Handling tools and equipment	Minor cuts/ injuries from handling	P	• Use tools correctly and return them to storage after use. • Avoid handling sharp components unnecessarily. • Locate the first aid kit and keep extra supplies on hand if needed.	Low
LiPo charging	Flammable/ a fire hazard	E	• Inspect the battery beforehand for any signs of swelling or a 'sweet' smell from leaking cells. • Monitor individual cell voltages before charging (should be above 3.8 V ideally) • Do not leave the charger unattended while charging. • Stop once cell voltages reach 4.2 V. • The battery will be placed in a fireproof bag while charging.	Medium

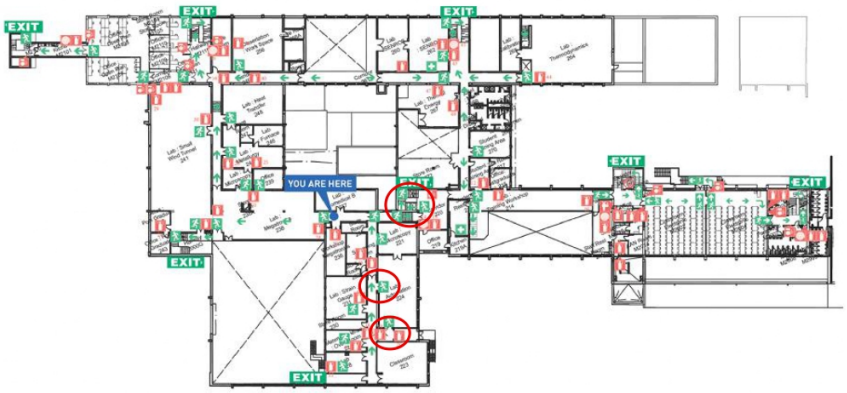
*P – personal, E – equipment

Disciplinary Actions

Failure to comply with any of the aforementioned safety regulations or procedures will result in disciplinary action. Students will be issued an initial warning: after three warnings, the lab access is revoked for a month.

Appendix A: Emergency Evacuation Plans

MECHANICAL & MECHATRONIC
ENGINEERING
Level 2





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